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Developed Markets
**North America**

# 2014 US Rates Outlook

## Beyond Base Camp

- **Yields Rise at a Contained Pace** — We expect Treasury yields to continue normalizing in 2014. However, we expect that while economic conditions will improve more than in 2013, Treasury yields will rise less and trade in a narrower range. We favor the belly of the curve despite apparent richness to recent history.

- **TIPS Hangover** — Real yields will move higher as an improving economy continues to reduce the attractiveness of low real yields in favor of riskier assets. Inflation is likely to run low in 2014 and pickup only gradually. However, breakevens are low relative to our expectations of inflation and are likely to move higher, especially in the front end. TIPS asset swaps are attractive.

- **Short-term Rates Remain Low** — Reserve creation coupled with modest reductions in short-term paper and potentially larger reductions in repo balance sheet imply another year of very low short-term interest rates. We think a cut in IOER remains unlikely.

- **Volatility Much Lower** — We expect vol to decline substantially next year on reduced economic uncertainty and reduced noise.

- **Swap Spreads Wider** — In 2014 we expect swap spreads to widen as the LIBOR – GC basis widens and deficits remain relatively low. We are skeptical that the SLR will have a significant tightening impact on swap spreads.

- **Basel III Regulation and US Rates** — The SLR is likely to be the most important new regulation for rates market in 2014 leading to increased Treasury price volatility and more persistent yield curve dislocations. We see the LCR as less important in 2014 given the large amount of Fed reserves on bank balance sheets.

- **Agency Debt** — 2014 will likely prove to be an ideal environment for callable investors from a rates, vol. and regulatory perspective.

- **Agency MBS** — We expect overseas investors and domestic banks to become net buyers in 2014 and drive demand for agency MBS. Going into the new year, we recommend a modest underweight on mortgages versus Treasuries.

US Rates Strategy

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See Appendix A-1 for Analyst Certification, Important Disclosures and non-US research analyst disclosures.

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# Summary of Views

Figure 1. Strategy Summary Table

| US Rates | View | Recommended Positions |
|---|---|---|
| Duration | We expect Treasury yields to move higher again in 2014, but at a more measured pace than 2013. We expect 10yr and longer maturities to rise by more than forwards and short maturities to rise by less than forwards. | Favor long positions in the belly of the curve (5yr and 7yr). Short positions in the back-end of the curve – 20yr+ maturities. |
| Yield Curve | We expect steepening from the 5yr part of the curve out to the 30y part of the curve. We also expect curvature to bow inward even further, especially in 1H-2014. | 5yr-10yr, 10yr-30yr, 7-30yr steepeners<br>Long belly 2yr–5yr–10yr fly, 2yr-7yr-10yr fly, 2yr-7yr-30yr fly |
| Swap Spreads | We expect swap spreads to move wider in 2014. We think front-end swap spreads may widen by 3-5bp as the LIBOR – GC basis widens. 10y swap spreads may widen by around 10bp as they are tight relative to fair-value based on the level of deficits. | None |
| Gamma | We expect realized vol to be subdued on account of lower expected economic volatility, as well as the fact that 10y rates are much closer to fair economic value than they were at the start of 2013. Consequently gamma implied volatility should be subdued as well. Modestly higher vol resulting from the Volcker rule or other regulation, as well as reduced Fed credibility should not change this fundamental picture. | None |
| Vega | As an extension to our gamma view, we would be bearish on intermediate volatility for the year. 1y10y vol should trade somewhere in a range from 70bp/annum to 85bp/annum. Both these levels are below current levels. We find long expiry volatility, as well as high strike skew in 2y2y to be modestly cheap. | Sell 1y10y 100bp wide Strangles<br>Buy 2y2y A+50 Payer delta hedged |
| Inflation | We expect real yields to rise. Breakevens are likely to move higher, especially in the front end | Long front end breakevens;<br>Long 10yr TIPs on asset swap |
| MBS | We expect overseas investors and domestic banks to become net buyers in 2014 and drive demand for agency MBS. Going into the new year, we recommend a modest underweight on mortgages versus Treasuries. | Modest underweight on MBS versus Treasuries<br>Short G2/FN 3.5s swap |
| Agency And SAS Debt | We expect agencies to deliver positive nominal returns in spite of some spread widening in 2014.<br><br>We are neutral on the SAS spreads. We don't expect much widening given strong investor demand. | On a relative basis we like 5yr bullet sector and 4nc6m callables in agencies.<br>Buy 5yr Renten, Sell 5yr KFW or<br>Buy 5yr FNMA, Sell 5yr KFW |

Source: Citi Research

# 2014 US Rates Outlook: Beyond Base Camp

**Brett Rose**

2014 is likely to be a year when we need to further analyze what 'normalization' means. Entering 2013 Treasury yields were driven more by tail risks, most notably European risk factors. Now they are driven primarily by US economic fundamentals and the associated expectations surrounding the Federal Reserve. Our expectation is that 2014 will feature an acceleration of the economic recovery and a continuation of the rise in Treasury yields (see Figure 2 for our economists' forecasts and Figure 4 for our Treasury yield forecasts). While we expect the economic recovery to be more dramatic in 2014 than it was in 2013, we think that the rise in Treasury yields will be more muted in 2014. This is because the rise in Treasury yields in 2013 went a long way towards realigning US Treasury yields with US economic data.

Figure 2. Economic Forecast Overview

| | 2013 | 2014 | 2015 |
|---|---|---|---|
| Real GDP 4Q/4Q Change | 2.0 | 3.1 | 3.0 |
| Core PCE Deflator | 1.3 | 1.8 | 2.0 |
| Unemployment Rate | 7.0 | 6.4 | 5.9 |

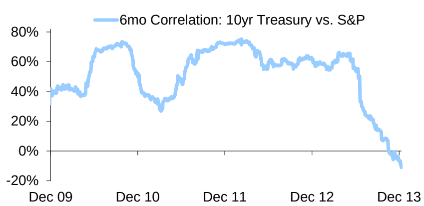
Source: Citi Research

## Bond Bubble Busted

While it is always difficult to precisely define what a bubble is, entering 2013 10yr TIPS real yields and estimated term premium of 10yr nominal Treasuries were both significantly negative. There may have been good reasons for this at the time, such as the Fed's QE purchases and the insurance value they provided to risk assets. However, it also seemed fairly evident that this was not a long-term sustainable level of valuations. Currently 10yr real yields and term premium are both positive; relative to last year 10yr real yields and term premium are about 150bp and 100bp higher respectively.

In May the combination of strong employment data and Fed official discussion of tapering of QE purchases significantly shifted expectations for the total likely amount of QE purchases and led to a greater than 100bp rise in 10yr Treasury yields by the end of June. During this May-June rise in Treasury yields the correlation between 10yr Treasury yields and S&P fell significantly and is now near zero (see Figure 3). This reduced the insurance, or hedging, value of Treasuries and we believe most Treasury positions that were being used in this way were sold in 2Q and 3Q 2013.

Figure 3. Treasury Hedge Value Breaks Down




Source: Bloomberg and Citi Research

We review this past behavior only in the interest of the implications to our future expectations. We think that the major impact of QE tapering has already occurred during discussions by Fed Governors in mid-2013 and that the market impact of the actual taper and ending of QE will be extremely modest. We also believe that the reduction of Treasury holdings as an insurance vehicle will result in less volatile moves higher in Treasury yields in 2014 than the middle of 2013. This drives our expectation that while we expect a stronger recovery in 2014 than in 2013, we expect it to result in a more muted rise in Treasury yields.

Figure 4. 2014 Treasury Yield Forecasts, Year-End and Mid-Year, Relative to Forward Curves

| | Current | YE Mkt Forward | YE Citi Forecast | Relative to Market | 2Q-14E Mkt Forward | 2Q-14 Citi Forecast | Relative to Market |
|---|---|---|---|---|---|---|---|
| 2yr Treasury | 0.30 | 0.93 | 0.85 | -0.08 | 0.60 | 0.50 | -0.10 |
| 5yr Treasury | 1.47 | 2.23 | 2.20 | -0.03 | 1.85 | 1.70 | -0.15 |
| 10yr Treasury | 2.81 | 3.21 | 3.30 | 0.09 | 3.03 | 3.05 | 0.02 |
| 30yr Treasury | 3.84 | 4.07 | 4.40 | 0.33 | 3.96 | 4.25 | 0.29 |
| | | | | | | | |
| 2-5 Curve | 1.17 | 1.30 | 1.35 | 0.05 | 1.25 | 1.20 | -0.05 |
| 5-10 Curve | 1.34 | 0.98 | 1.10 | 0.12 | 1.18 | 1.35 | 0.17 |
| 10-30 Curve | 1.03 | 0.86 | 1.10 | 0.24 | 0.93 | 1.20 | 0.27 |
| 2-5-10 Fly | -0.17 | 0.32 | 0.25 | -0.07 | 0.07 | -0.15 | -0.22 |
| 5-10-30 Fly | 0.31 | 0.12 | 0.00 | -0.12 | 0.25 | 0.15 | -0.10 |

Source: Citi Research

## Quantifying Normalization of Treasury Yields

Given our expectations for a more robust economic recovery in 2014, we need to assume that Treasury yields substantially normalize. The question to ask is, what does normalization mean following a 30-year drop in 10yr Treasury yields from above 15% in 1981 to below 1.5% in 2012? While it may seem obvious that Treasury yields are not likely to reverse that multi-decade drop, we need to have some objective way to quantify how much of a reversal is reasonable. To do this we look at long term estimates of term premium and long-term relationships between Treasury yields and nominal GDP growth.

### Term Premium Approaching 'Normal Levels'

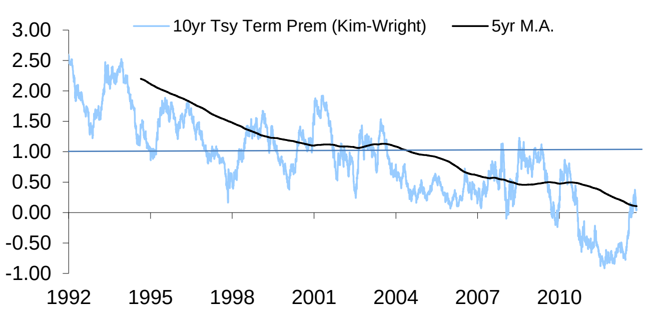
**10yr term premium is only 25bp below the decade average prior to the crisis**

To look at the estimated term premium over a long period of time we use the data provided by the Federal Reserve. In Figure 5 we show that the current term premium level is about 25bp, while the longer term average is about 100bp. However, the steady decline of term premium over the time period may reflect that increased credibility of the Federal Reserve has lowered required term premium. The average of just over 50bp in the decade prior to the crisis is probably a more appropriate expectation for the current environment. This would suggest that 10yr Treasury yields now should be about 3.10%.

We can also crudely estimate term premium by mapping out specific Fed Funds paths and comparing these paths with various points on the Treasury curve. Using this method we are able to closely approximate the Kim-Wright results in recent periods using reasonable Fed Funds paths. The advantage of this is that it allows us to stress the Fed Funds paths. For example, we can match the Kim-Wright term premium for 10yr using the following fed Funds path: December 2015 initial hike of Fed Funds, a 100bp/yr hike pace and a terminal Fed Funds rate of 4%. This is relatively consistent with our base assumption – although we expect the initial hike in 3Q-2015.

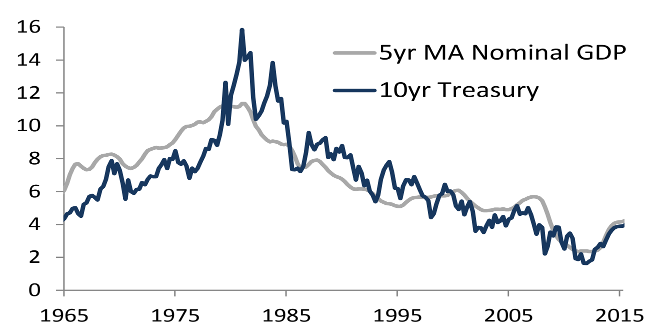
To estimate a fair value of the 10yr, 1yr forward we take the Fed Funds path from above and shift the starting date 1yr forward. Doing this suggest that 10yr Treasury yields should be 31bp higher 1-year in the future, relative to today. Based on the 3.10% current 10yr estimate, 3.41% would be the estimate for the end of 2014. This is very close to the 3.30% year-end target in Figure 4 above.

Figure 5. Term Premium May Tend Towards 50-100bp




Source: Federal Reserve and Citi Research

Figure 6. 10yr Treasury Yields May Tend Towards 4% at Cycle Top




Source: Bloomberg and Citi Research

### Nominal GDP Expectations and 10yr Treasury Yields

Another way to get an appropriate 'terminal' level on 10yr Treasury yields is to look at a very long-term relationship between nominal GDP and 10yr Treasury yields (see Figure 6). Over the past 50-years 10yr Treasury yields have averaged about

25bp less than the 5yr moving average of nominal GDP. Based on our longer term economic expectations this would suggest a top of cycle 10yr Treasury yield of just above 4%. We do not expect 2014 to be the top of the cycle (Fed Funds are likely to remain on hold the entire year), so 4% as a mid-range point on 10yr yields is probably still a few years away. However, we do expect 10yr yields to move safely above 3% in 2014 and end the year at 3.3%

## Trading on a Curve

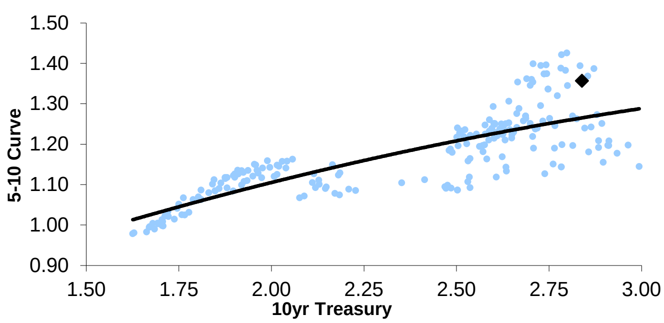
**The 5yr – 10yr Treasury curve will likely bear flatten in 2014 as the initial Fed Funds date draws near. However, further steepening is likely early in 2014.**

Perhaps even more interesting in 2014 will be the evolution of the Treasury curve. We expect the Federal Reserve will end QE 3 by 3Q-2014, with the initial tapering of purchases coming in 1Q-2014. Since QE 3 purchases focused on longer maturity securities and one of the main impacts of QE thought to be on term premium, this should have the most impact on longer maturity yields. While we think that the taper is substantially priced in, the flow impact may be felt the most in the very back of the curve where liquidity is the lowest.

The actual impact of QE appears to have been far more complicated. Most often changing QE expectations have often equally impacted shorter-maturity yields as Fed Fund expectations have moved in parallel. This is likely due to the perceived signaling value of QE. This something the Federal Reserve expressed frustration in during 2Q-2013 when 'taper talk' had the most impact on the 5yr and 7yr part of the curve. Since then, the Fed has been diligent in trying to disentangle QE expectations and Fed Funds expectations - with a degree of success finally in late 2013 following the publication of two Fed staff papers. Therefore, the curve looks to be too steep relative to history (see 5yr – 10yr curve in Figure 7).

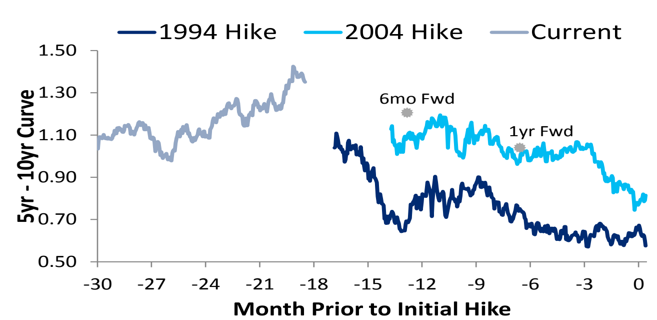
Despite this steeping near year-end, we think that room remains for additional steepening – especially relative to forwards. One way to look at this is to compare curve levels relative to the amount of time to the first hike of Fed Funds. We assume that the initial hike takes place in 3Q-2015 and compare with the hiking cycles in 1994 and 2004. Since the Fed was not on hold for more than 18-months prior to the initial hike in 1994 or 2004, we use current forwards to overlap with these curves. Figure 8 shows that current market levels of the 5yr – 10yr curve are in line with the 2004 Fed tightening cycle, when adjusted for months prior to initial hike.

Figure 7. 5yr – 10yr Curve Steep to Recent History




Source: Bloomberg and Citi Research

Figure 8. But Not Necessarily to Mid-2015 Hike Date



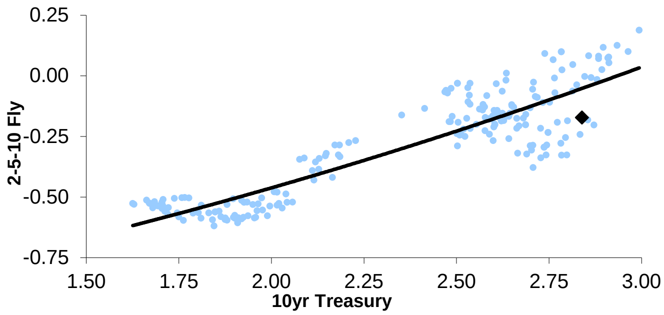

Source: Bloomberg and Citi Research

Given the potential for a flatter path of hikes in this cycle, perhaps this curve should be even steeper early in 2014. However, by late 2014 this curve risks flattening dramatically as the date of the initial hike draws nearer.

### Similar Story in Curvature

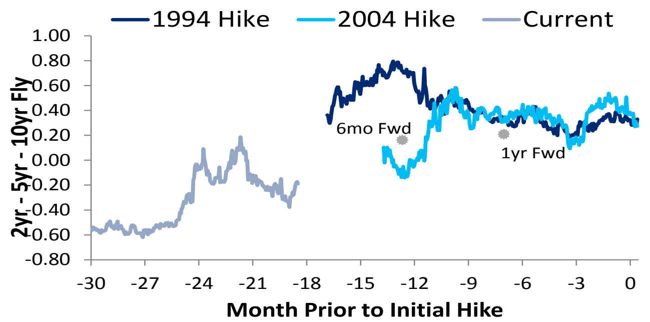
The dynamic looks very similar when looking at curvature, with the belly of the curve looking rich to the wings (see Figure 9). Again, this shift took place in November when the 5yr fly outperformed even as the yield level of Treasuries drifted higher. However, if we compare the current forwards relative to similar points in the 1994 and 2004 hiking cycle, the 5yr actually appears slightly cheap (see Figure 10).

Figure 9. 2yr - 5yr – 10yr Fly Also Rich to Recent History




Source: Citi Research

Figure 10. But Also in Line to Mid-2015 Hike Date




Source: Citi Research

The strong carry of the 5yr fly also makes it an attractive position – especially given our expectation of a falling volatility environment in 2014 (see the Volatility section for more details on this forecast).

### 10yr – 30yr Curve

We continue to expect the back of the curve to perform poorly. This is driven by a variety of factors, including:

- Expectations for falling volatility, which provides a smaller convexity value to the back-end of the curve.
- Rising inflation expectations, which decreases value of longer maturities.
- The wind-down of QE, which should hurt the least liquid parts of the curve most, even though we do not expect the actual flow to have a meaningful impact on the level of interest rates.
- Increased concentration of Treasury securities outstanding at the very back end of the curve.

Increased insurance / pension demand as yields levels move towards new highs is one likely counterpoint to our view. However, despite this final point, we continue to favor 10yr – 30yr steepeners. We expect this curve to move higher in level, despite the significant flattening built into the forward curve.

## Summary

Our expectation is that Treasury yields will continue to move higher through 2014, albeit at a slower pace than in 2013. Although we think the economic recovery in 2014 is likely to be more convincing in 2014 than it was in 2013, we think that much of the normalization rates was already accomplished in mid-2013. Therefore, we think that a rise to 3.30% in 10yr Treasury yield seems reasonable by year-end 2014. Despite the richness of the belly of the curve (5yr / 7yr) relative to recent history, we think that this part of the curve will perform the best – at least in the first half of 2014. We would recommend the following positioning:

- Long positions focused in the belly (5yr and 7yr) of the curve.
- Short positions in the back end of the curve (20yr and 30yr)
- Neutral in 2yr, 3yr, 10yr.
- Steepeners (5yr-10yr, 7yr-10yr, 10yr-30yr)
- Long the belly of the fly (2yr-5yr-10yr. 2yr-7yr-10yr, 2yr-7yr-30yr)

This series of recommendations is consistent with a fairly tight range of yields and falling volatility. The major risk scenario could be hedged by buying out-of-the money payers on the front-end of the curve (6m3y, for example). However, we think our expectation for a tight trading range is reasonable given that rates have substantially normalized to economic data and traditional Fed policy (the Fed Funds rate) is likely to be constant for the entire calendar year.

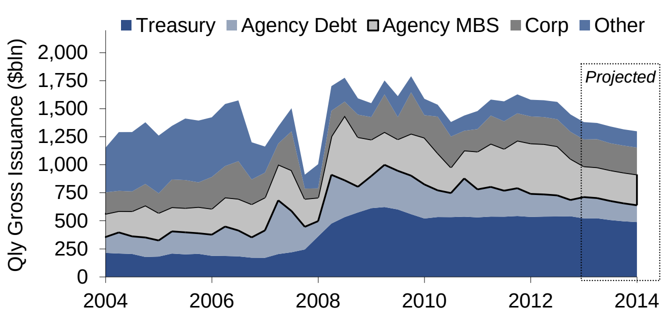
# Supply & Demand

**Brett Rose**
**Jabaz Mathai**
**Andrew Hollenhorst**

Overall fixed-income supply should fall by about 10% in 2014 relative to 2013. Most sectors, including US Treasuries, should be down modestly. The outliers should be Agency MBS which should fall more aggressively, and Corporates, which should rise. In Figure 11 we show quarterly gross fixed-income issuance since 2004.

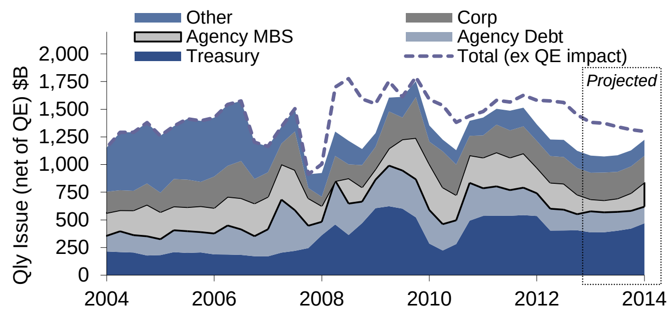
If we exclude net Federal Reserve purchases, however, supply looks very different. Doing that, gross supply to the private market is expected to be close to flat. In Figure 12 we show quarterly gross fixed-income issuance net of Federal Reserve purchases since 2004.

Figure 11. Gross US Fixed-Income Supply Falling in 2014




Source: US Treasury, embs, Bloomberg and Citi Research

Figure 12. Issuance Flat Net of QE




Source: US Treasury, Federal Reserve, embs, Bloomberg and Citi Research

The above figures also illumistrate that despite the dramatic rise in US Treasury issuance since 2008, the issuance in the overall US fixed income market has remained relatively constant. The starts and stops of the various QE programs have had the biggest impact on the true supply of fixed-income to the public market.

## Treasury Funding Needs Falling, Net More Than Gross

In Figure 13 we show that Treasury net supply is expected to continue falling in 2014 – as it has each year since 2009. However, despite the near 60% drop in net funding needs since 2009, gross supply of coupon Treasuries has remained remarkably constant. We expect this same trend to continue in 2014.

**Net funding needs have fallen dramatically since the peak in 2009. However, gross issuance has fallen very modestly and is expected to continue doing so given rising redemptions and the impact of falling T-bills.**

Figure 13. The Drop in Gross Supply Fairly Modest Relative to Net Supply

| | Gross Supply Cpn & TIPS | Redeem Cpns &TIPS | Net Supply Cpns &TIPS | Net Supply T-Bills | Net Supply FRNs |
|---|---|---|---|---|---|
| 2007 | 581 | 497 | 84 | 60 | |
| 2008 | 922 | 532 | 390 | 863 | |
| 2009 | 2,109 | 542 | 1,568 | -73 | |
| 2010 | 2,247 | 776 | 1,472 | -21 | |
| 2011 | 2,135 | 766 | 1,370 | -252 | |
| 2012 | 2,153 | 1,172 | 981 | 108 | |
| 2013 (forecast) | 2,140 | 1,246 | 894 | -41 | |
| 2014 (forecast) | 2,015 | 1,418 | 597 | -106 | 135 |

Source: US Treasury and Citi Research

Note: We are reporting on a calendar year basis, not a fiscal year basis.

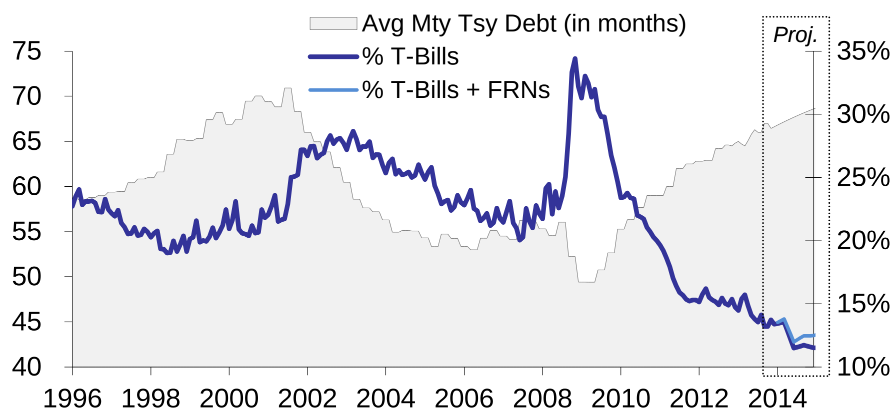
Treasury auction schedules (excluding T-Bills) have remained extremely consistent since 2010. The modest reduction in the size of 2yr and 3yr auctions in the second half of 2013 were the first changes to a nominal coupon auction since the end of

2010. However, even these reductions have moved at a more moderate pace than we had expected. While we expect some additional reductions in the size of shorter maturity auctions in 2014 we think it is important to understand why the dramatic drop in deficit and funding needs have only resulted in modest reduction in the size of coupon auctions.

- The initial rise in funding needs (in 2008) was met primarily by a sharp increase in T-bill issuance. T-Bill net issuance has been cumulatively negative since 2009 and likely will be near zero in 2013 and 2014. This mutes the volatility of net funding needs met by coupon Treasuries.
- The dramatic rise in Treasury debt outstanding results in a dramatic rise in debt that needs to be rolled over. Therefore, while net Treasury funding needs satisfied by coupon issuance is falling significantly, the amount of coupon issuance needed to replace maturing issues is rising significantly.
- While deficit projections for the next several years look favorable, longer-term deficit projections do not – which will likely result in the need to raise coupon auction sizes in the longer-term
- The short-term nature of recent budget deals have resulted in great uncertainty surrounds medium term funding needs and maintaining larger auction sizes is the more conservative way to address this uncertainty.

Underlying all of this is a desire by the US Treasury to increase the average maturity of Treasury debt outstanding. Therefore, the Treasury will likely continue to err on the side of maintaining larger coupon auction sizes and address any slack in funding with reductions in T-Bills outstanding (see Figure 14).

Figure 14. The US Treasury Has Continued to Increase the Average Maturity of Treasury Debt




Source: US Treasury and Citi Research

## US Treasury Funding Strategy Changes Expected in 2014

- We expect the size of 2yr, 3yr and 5yr auctions to be gradually reduced over the course of 2014. The levels we forecast for the end of 2014 represent a likely long-term low in coupon auction sizes that we see as feasible. Therefore, we would not be surprised if reductions were more modest.
  - Aggregate monthly issuance of 2yr/3yr/5yr fell from $102 bln/mo at the beginning of 2013 to $97 bln/mo at the end of 2013. We expect that it will fall to only $83 bln/mo by the end of 2014.
- We expect a shift to smaller, more frequent 5yr TIPS auctions during 2014. The Treasury noted in the November refunding announcement that it is studying such an idea. We expect more details to be included in the February refunding announcement.
  - We expect that the Treasury will issue a new 5yr TIPS in October with a reopening of that issue in December. The April 5yr TIPS would only be reopened once.
  - This would increase the total number of TIPS auctions from 12 to 13, with October having both a 5yr and a 30yr auction. While we are fairly confident of an outcome similar to this in principle, the precise outcome could be very different.

**FRNs will be the first new product that the US Treasury has introduced since TIPS over 15 years ago.**

- The Treasury will initiate a Floating Rate Note (FRN) program in 2014, with its initial auction on January 29. FRNs should be considered as a substitute for T-bills. We highlight the broad characteristics of FRNs below, with further details available in the offering circular.
  - Auctions are expected to be $10 billion in size and potentially grow to $15 billion later in 2014. We think Treasury will set larger new-issue auction sizes (e.g. $15 billion) with smaller reopenings (e.g. $10 billion). FRNs will accrue interest daily and pay interest quarterly. $120 - $140 billion in issuance in 2014 seems reasonable.
  - FRN auctions will be monthly, with a quarterly new issue and two re-openings. New issues will be at the end of Jan., Apr., Jul. and Oct. and the on-the-run issue will be reopened at the end of each of the other 8 months. See the current tentative US Treasury auction schedule.
  - The maturity will be 2yrs, with other maturities possible in the future.
  - FRNs will be a daily reset floater indexed to the most recent 13-week T-bill auction rate (currently only weekly). This means the duration of FRNs will be close to zero duration.[1]
  - We expect pricing to be approximately 5-10bp above T-bills. See our November article for more complete pricing considerations.

Note that all of these changes are consistent with the US Treasuries desire to continue increasing the average maturity of Treasury shown in Figure 14 above.

[1] The fact that FRNs are daily reset, even though 13-week T-bill auctions are currently only weekly is an important technical distinction as it allows MMF to count FRNs as 1-day instead of 7-day securities for the purpose of WAM calculations. Under rule 2a-7 MMF must keep portfolio WAM under 60 days.

## Treasury Coupon Auction Expectations for 2014

Below we show our expectations for Treasury coupon auctions in 2014 (Figure 15) and 2013 (Figure 16) for comparison.

Figure 15. Expected Treasury Coupon Supply – 2014

| | 2yr | 3yr | 5yr | 7yr | 10yr | 30yr | 5yr TIPS | 10yr TIPS | 30yr TIPS | Gross Supply | Redeem | Net Supply |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Jan-14 | 32 | 30 | 35 | 29 | 21 | 13 | | 15 | | 175 | 118 | 57 |
| Feb-14 | 32 | 30 | 35 | 29 | 24 | 16 | | | 9 | 175 | 126 | 49 |
| Mar-14 | 32 | 30 | 35 | 29 | 21 | 13 | | 13 | | 173 | 101 | 72 |
| Apr-14 | 30 | 28 | 35 | 29 | 21 | 13 | 13 | | | 169 | 117 | 52 |
| May-14 | 30 | 28 | 35 | 29 | 24 | 16 | | 13 | | 175 | 132 | 43 |
| Jun-14 | 30 | 28 | 35 | 29 | 21 | 13 | | | 7 | 163 | 104 | 59 |
| Jul-14 | 28 | 26 | 33 | 29 | 21 | 13 | | 15 | | 165 | 125 | 40 |
| Aug-14 | 28 | 26 | 33 | 29 | 24 | 16 | 12 | | | 168 | 134 | 34 |
| Sep-14 | 28 | 26 | 33 | 29 | 21 | 13 | | 13 | | 163 | 107 | 56 |
| Oct-14 | 25 | 26 | 32 | 29 | 21 | 13 | 13 | | 7 | 166 | 108 | 58 |
| Nov-14 | 25 | 26 | 32 | 29 | 24 | 16 | | 13 | | 165 | 137 | 28 |
| Dec-14 | 25 | 26 | 32 | 29 | 21 | 13 | 12 | | | 158 | 109 | 49 |
| Total | 345 | 330 | 405 | 348 | 264 | 168 | 50 | 82 | 23 | 2,015 | 1,418 | 597 |
| *vs. 2013* | -63 | -47 | -15 | 0 | 0 | 0 | 0 | 0 | 0 | -125 | 172 | -297 |

Source: Citi Research

Figure 16. Actual Treasury Coupon Supply – 2013 (note, expected for December)

| | 2yr | 3yr | 5yr | 7yr | 10yr | 30yr | 5yr TIPS | 10yr TIPS | 30yr TIPS | Gross Supply | Redeem | Net Supply |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Jan-13 | 35 | 32 | 35 | 29 | 21 | 13 | | 15 | | 180 | 89 | 91 |
| Feb-13 | 35 | 32 | 35 | 29 | 24 | 16 | | | 9 | 180 | 109 | 71 |
| Mar-13 | 35 | 32 | 35 | 29 | 21 | 13 | | 13 | | 178 | 93 | 85 |
| Apr-13 | 35 | 32 | 35 | 29 | 21 | 13 | 18 | | | 183 | 108 | 75 |
| May-13 | 35 | 32 | 35 | 29 | 24 | 16 | | 13 | | 184 | 110 | 74 |
| Jun-13 | 35 | 32 | 35 | 29 | 21 | 13 | | | 7 | 172 | 91 | 81 |
| Jul-13 | 35 | 32 | 35 | 29 | 21 | 13 | | 15 | | 180 | 111 | 69 |
| Aug-13 | 34 | 32 | 35 | 29 | 24 | 16 | 16 | | | 187 | 135 | 51 |
| Sep-13 | 33 | 31 | 35 | 29 | 21 | 13 | | 13 | | 178 | 92 | 83 |
| Oct-13 | 32 | 30 | 35 | 29 | 21 | 13 | | | 7 | 166 | 91 | 76 |
| Nov-13 | 32 | 30 | 35 | 29 | 24 | 16 | | 13 | | 178 | 122 | 57 |
| Dec-13 | 32 | 30 | 35 | 29 | 21 | 13 | 16 | | | 175 | 95 | 81 |
| Total | 408 | 377 | 420 | 348 | 264 | 168 | 50 | 82 | 23 | 2,140 | 1,246 | 894 |

Source: US Treasury and Citi Research

### T-bill Issuance Volatility and the Debt Ceiling

**T-bill needs will be very low in early 2014. Longer term pattern forecasts are difficult given debt ceiling episodes.**

T-bill issuance always exhibits seasonal volatility as it is the buffer for seasonal cash need differences given tax collection patterns. However, this has been exacerbated in recent years on several occasions due to debt ceiling considerations. Further, lumpy TARP proceeds and GSE dividends have also been significant considerations. Early 2014 will be impacted by $39 billion in GSE dividends at year-end, light seasonal cash needs in January and then the need to reduce cash in February due to debt ceiling suspension that expires on February 7. Unfortunately, ongoing mini fiscal crises will probably continue to make longer term T-bill funding forecasts difficult.

## Treasury Demand: Fed Slowly Replaced by Foreign in 2H

In Figure 17 we show our estimates for ownership of Treasuries over the past several years.[2] We see the top three as particularly important to follow as the Federal Reserve and Non US investors own approximately 70% of US Treasuries and the household sector can be highly volatile in its ownership levels.

Figure 17. Estimated Changes in Treasury Ownership

| | Level of Ownership (in $'s bln) | | | | Growth in Ownership (in $'s bln) | | | Growth in Ownership (in %) | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 2011 | 2012 | 2013 | 2014 | 2012 | 2013 | 2014 | 2012 | 2013 | 2014 |
| Federal Reserve | 1,645 | 1,666 | 2,206 | 2,476 | 21 | 540 | 270 | 1% | 32% | 12% |
| Non US Investors | 4,359 | 4,910 | 5,128 | 5,385 | 551 | 219 | 256 | 13% | 4% | 5% |
| Households | 419 | 606 | 378 | 339 | 187 | (228) | (39) | 44% | -38% | -10% |
| Banks, S&L's | 224 | 300 | 241 | 246 | 77 | (59) | 5 | 34% | -20% | 2% |
| Insurance & Pension | 792 | 896 | 935 | 981 | 104 | 39 | 47 | 13% | 4% | 5% |
| Mutual Funds | 535 | 627 | 691 | 708 | 92 | 64 | 17 | 17% | 10% | 3% |
| All Other | 750 | 755 | 694 | 736 | 6 | (61) | 42 | 1% | -8% | 6% |

Source: Federal Reserve and Citi Research

This table shows the Federal Reserve dominated net demand in 2013, unlike 2012 when net demand was more broadly distributed. 2014 is likely to begin where 2013 ended, but transition to something more similar to 2012 in the second half of the year.

### Fed Will Continue to Dominate Demand in 1H-2014

The Federal Reserve's QE program purchased almost 60% of net Treasury supply and 25% of gross Treasury supply in 2013. These ratios are likely to remain fairly constant in 1H-2014 as well. However, assuming the QE program is phased out over the middle quarters of 2014 as we expect, the Fed will be a much more modest contributor to Treasury demand in 2H-2014 and beyond. Note that the Fed's share of gross purchases is even greater when considered in duration terms due to its skewing of purchases to longer maturity issues.

Figure 18. The Fed's Domination of Treasury Issuance Will Likely Fade as 2014 Goes on

| | --- 2013 --- | | --- 2014, 1H --- | | --- 2014, 2H --- | |
|---|---|---|---|---|---|---|
| 2013 | Par | Dur in 10s | Par | Dur in 10s | Par | Dur in 10s |
| Fed SOMA Purchases of Treasury Securities | 540 | 621 | 225 | 259 | 45 | 52 |
| Gross Treasury Supply | 2,140 | 1,432 | 1,030 | 703 | 985 | 695 |
| Net Treasury Supply | 894 | 1,432 | 332 | 703 | 265 | 695 |
| Fed % Of Treasury Gross Issuance | 25% | 43% | 22% | 37% | 5% | 7% |
| Fed % Of Treasury Net Issuance | 60% | 43% | 68% | 37% | 17% | 7% |

Source: US Treasury, Federal Reserve and Citi Research

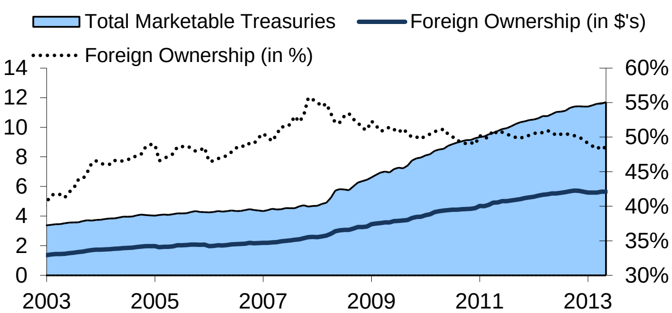
[2] This is an estimate of ownership of Treasury coupon securities (including TIPS) and does not include Treasury Bills. Actual data is based on the Federal Reserve's Flow of Funds, but makes estimates to exclude T-bills which are only specifically provided for certain investor classes. We also combine groups to minimize the number of investor classes.

### Non US Investors are Likely to Continue Buying - at a Slower Pace

The most obvious candidate to replace the Federal Reserve when it ultimately ends its QE purchases is foreign investors – if only because, other than the Fed, they are by far the largest participant in the US Treasury market. Figure 19 shows that the pace of foreign ownership has moderated in 2013. However, this is almost by definition given that the Fed has purchased almost 60% of net issuance in 2013. We expect foreign ownership to rise from a 2% growth rate in 2013 to a 5% growth rate in 2014. This includes a near double digit growth rate in 2H-2014, which while high, is still below levels from 2012.

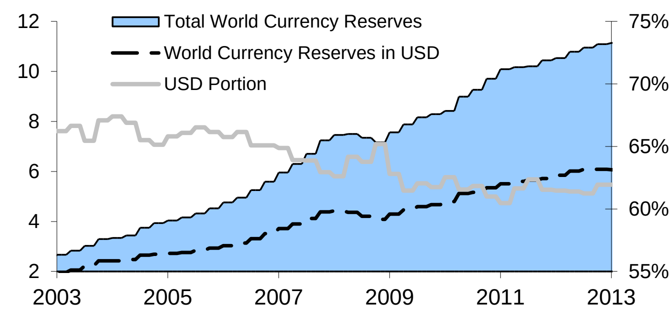
Currency reserve growth has been moderating recently and anecdotally there has been discussion that China may be seeing reduced benefits from further rises in its currency reserves. In Figure 20 we show that currency reserves continue to rise – albeit at a reduced pace. While the percentage of world currency reserves held in USD has been declining for many years, the pace of decline remains extremely modest.

Figure 19. Foreign Demand Has Moderated During QE Periods




Source: US Treasury, Federal Reserve and Citi Research

Figure 20. Currency Reserves Still Flock to USD

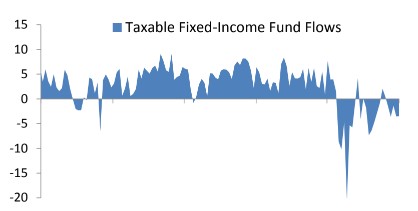



Source: Bloomberg and Citi Research

### Households And Funds Likely to Reverse Course Mid-Year

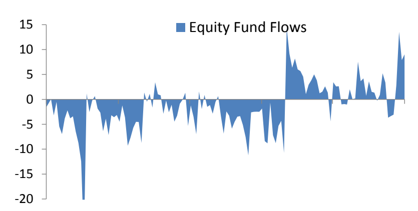
Households were significant net sellers of US Treasuries in 2013 which we attribute to an asset allocation decision to move more into equities. Mutual fund flows are likely reacting to the same performance discrepancy. Following the New Year's Eve resolution of the debt ceiling, flows into equities started in earnest (see Figure 22), Fixed-Income flows then flipped negative following the Fed's initial 'taper talk' in May (see Figure 21).

Figure 21. FI Fund Flows Flipped in May




Source: Haver, ICI and Citi Research

Figure 22. Equity Fund Flows Flipped Earlier




Source: Haver, ICI and Citi Research

We would expect these types of flows to remain in place for households and mutual funds past the initial tapering of QE. However, we expect them to moderate should the reduction of Fed purchases occur uneventfully – as we expect it will. Overall, we expect a modest drop in household and mutual fund ownership of Treasury securities in 2014.

### Banking Purchases Should Level Out in 2H-2014

Bank ownership of Treasury securities fell by 20% in 2013. Given LCR we expect Treasury and Ginnie Mae securities to grow as a percentage of bank portfolios going forward. However, with cash on balance sheets being dominated by Fed reserves, growth of aggregate securities portfolios is likely to be negative until the end of QE 3 purchases. Overall, we expect modest growth in 2014. The substantial cross-winds of evolving bank regulation make this the most difficult investor category to forecast. However, we feel that banks have substantially prepared for regulations already and that most impacts have been more modest than initial concerns. Therefore, we are comfortable expecting a modest drop in 1H-2014 which is offset by a modest rise in 2H-2014.

## Rising Equity Prices Reduce Rationale for Great Rotation

Investors (and analysts) that specialize in specific asset classes can miss major trends that cut across asset classes. We suspect that much of the gap higher in Treasury yields in May-June 2013 was related to equity investors. This likely included both levered equity investors that were using fixed-income as a hedge product and unlevered investors that make asset allocation decisions between equities and fixed-income. Our suspicion is that levered investors have substantially removed fixed-income as a hedge against equity positions since correlations are no longer as supportive. However, it is less clear how much of the likely shift in asset allocation has been accomplished.

One way to analyze this is to look at the equity weighting of total assets of major groups that are reported in the Federal Reserve Flow of Funds. In Figure 23 we show that households, pension and insurance companies all have average or above-average equity weightings. However, it also shows that each are well below levels reached at the previous peak of the equity market in early 2000.

Figure 23. Current Equity Weightings in Line With Long-Run Average

| | Current (3Q-2013) | Average since 1980 | 1Q-2000 |
|---|---|---|---|
| Households | 19% | 16% | 28% |
| Pension | 30% | 30% | 43% |
| Insurance | 19% | 18% | 30% |

Source: Federal Reserve Flow of Funds and Citi Research

Our suspicion is that risk exposed in the aftermath of the 2000 equity market peak and the more recent great recession will likely prevent investors from moving back to 2000 equity market weightings. However, we would not be surprised if levels were to move higher than current levels. This suggests that the current pace of shift from fixed-income to equities could continue for at least part of 2014.

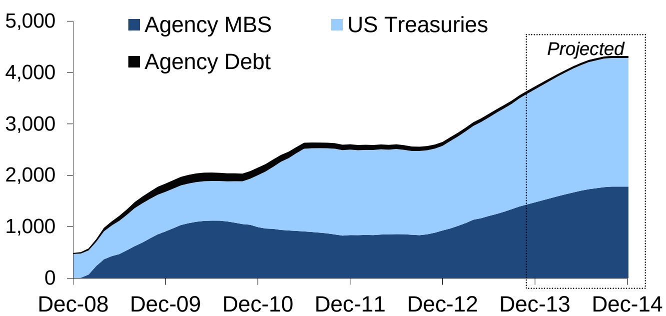
# Federal Balance Sheet and Policy Evolution

**Brett Rose**

2014 looks to be a transition period for the Federal Reserve's LSAP program. We anticipate that the pace of purchases will be reduced early in the year and come to an end before year-end. However, we expect that this will still result in almost $600 billion in net purchases in 2014. This will increase the total SOMA portfolio from $3.7 trillion at the end of 2013 to $4.3 trillion at the end of 2014. Therefore, the Federal Reserve will likely still be the most significant market participant in both the Treasury and the agency MBS markets in the first half of 2014. In Figure 24 we show the likely path of the SOMA portfolio in 2014.

The Fed's footprint in the Treasury and Agency MBS market has been growing in not only dollar amount, but also percentage of the market. This has been even more so in the Agency MBS market than the Treasury market due to much smaller net issuance in agency MBS relative to US Treasuries. In Figure 25 we show the percentage of the aggregate US Treasury and Agency MBS universe that the Fed owns. The Fed reached the highest levels of the past five years towards the end of 2013 and should move higher still by the expected peak in late 2014.

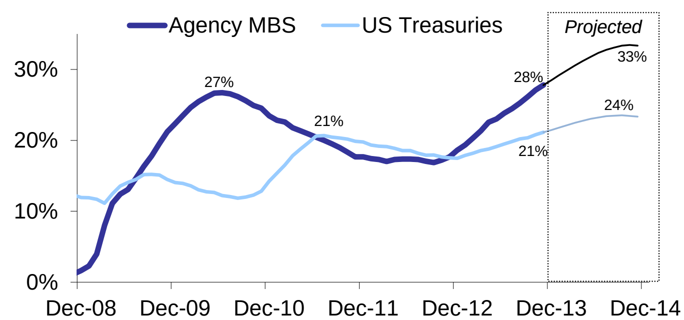
Figure 24. Fed Securities Portfolio Likely About $4.3 trillion at the Peak




Source: Federal Reserve, US Treasury, Bloomberg and Citi Research

Note: For the universe, we assume only fixed-rate agency MBS for the MBS universe and only coupon-paying marketable debt for the Treasury universe. Also, in Figure 24 we include the Treasury's ownership of MBS which peaked at about 4% of all MBS at the end of 2009 and is now zero. We do not include this in Figure 25.

Figure 25. Ownership Concentration Making New Highs in 2014




Source: Federal Reserve, US Treasury, Bloomberg and Citi Research

Note: Percentage ownership is par value weighted.

We do not anticipate these levels of ownership causing any deterioration of market function. This is despite our view that market function was disrupted in the agency MBS market in 2010 when overall concentration was lower. However, to more significantly analyze this we find it important to look at concentration levels by more granular market segment. Current Fed ownership in Agency MBS is more widely dispersed across the MBS universe than in 2010 – a wider selection of coupons. US Treasury ownership, however, is much less dispersed – a tighter selection of maturities. However, the highest concentration of US Treasury maturities now (shown in Figure 28) is lower than the most concentrated MBS coupons in 2010. This provides us with comfort that the Fed is not that close to impeding market function under QE 3.

### Net Supply Net of Fed to Remain Low

Even under our assumed QE path, with taper in early 2014 and conclusion in 3Q 2014, the Federal Reserve will still be purchasing a substantial amount of the net issuance across the Treasury and Agency MBS market. In Figure 26 we show the size of assumed Fed purchases relative to our estimates of gross and net supply – both for US Treasuries and agency MBS.

Figure 26. The Fed Continues to Dominate Net Issuance

| US Treasuries | Gross Issuance | Net Issuance | Fed Net Purchases | Net Issuance Net of Fed |
|---|---|---|---|---|
| 2012 | 2,153 | 981 | 21 | 960 |
| 2013 | 2,140 | 894 | 540 | 354 |
| 2014 | 2,015 | 597 | 270 | 327 |

| Agency MBS | Gross Issuance | Net Issuance | Fed Net Purchases | Net Issuance Net of Fed |
|---|---|---|---|---|
| 2012 | 1,601 | 96 | 120 | -24 |
| 2013 | 1,550 | 202 | 480 | -278 |
| 2014 | 1,140 | 180 | 305 | -125 |

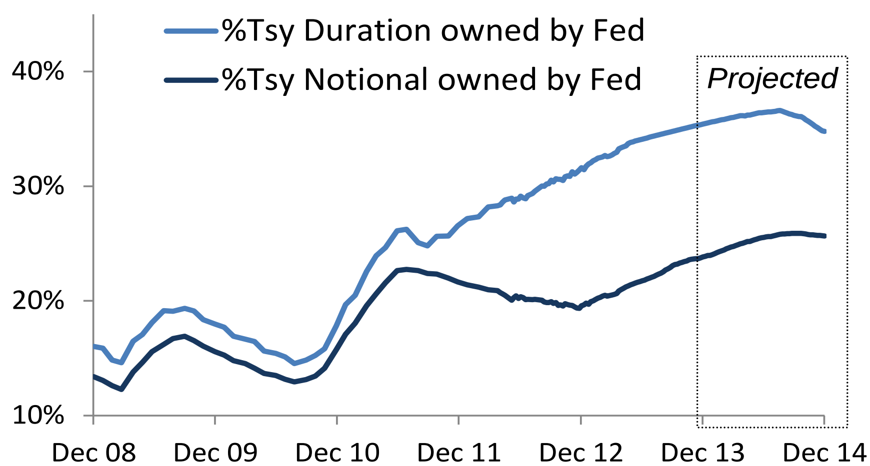
Source: Federal Reserve, US Treasury, embs and Citi Research

There is likely to be a significant difference in behavior between the first and second half of the year. In the first half of the year aggregate net-net issuance is likely to be slightly negative and in the second half of the year it is likely to be near $200 billion.

### In Duration Terms, Fed Impact is Even More Dramatic

We noted in Figure 25 that current ownership of the Treasury market is slightly higher than that at the end of QE 2 in mid-2011. However, given the focus on duration of purchases in Operation Twist and QE3, the amount of duration that the Fed owns is even more dramatic than the notional amount of securities. Figure 27 shows that the percentage of Treasury market duration that the Fed owns is far greater now and should reach 37% in 2014 compared with about 26% at the end of QE 2.

Figure 27. Notional Ownership is Near One-Quarter, Duration Ownership is Above One-Third




Source: Federal Reserve, US Treasury and Citi Research

### Higher Ownership Concentration in Certain Sectors

Over the course of Operation Twist and QE 3 the Federal Reserve has concentrated its purchases in longer term securities, so the overall concentration level shown in Figure 27 does not fully represent ownership concentration.  In Figure 28 we show that:

- Current concentrations levels are more pronounced than at the end of QE 2 (prior to Operation Twist and QE 3) when purchases were more evenly spread across the curve.
- Concentrations are most acute in 10yr and greater maturity buckets. This will continue to rise into mid-year, but then fall as QE purchases dwindle in the second half of 2014.
- By the end of 2014 concentration in the 3-4yr bucket will be relatively high as many of the purchases 4yr - 7yr maturity purchases from the last couple of years roll down in maturity

Figure 28. Current Concentrations Very High in Longer Maturity Treasury Securities

| | 7/31/2011 | 12/31/2013 | 6/30/2014 | 12/31/2014 |
|---|---|---|---|---|
| 0-3yr Maturity | 16% | 5% | 7% | 9% |
| 3-4 | 16% | 20% | 29% | 39% |
| 4-5 | 30% | 36% | 35% | 32% |
| 5-6 | 29% | 43% | 42% | 35% |
| 6-8 | 37% | 36% | 37% | 36% |
| 8-10 | 21% | 35% | 30% | 20% |
| 10-20 | 31% | 46% | 49% | 49% |
| 20-28 | 23% | 57% | 60% | 57% |
| 28-30 | 10% | 22% | 21% | 16% |
| TIPS | 11% | 10% | 11% | 10% |
| Bills + FRNs | 1% | 0% | 0% | 0% |
| Total | 17% | 19% | 20% | 20% |
| Total (ex Bills, FRNs) | 18% | 21% | 23% | 23% |
| Total (ex Bills, FRNs & TIPS) | 18% | 23% | 24% | 24% |

Source: US Treasury, Federal Reserve and Citi Research

It is also interesting to consider the combined impact of changing Treasury issuance patterns and Federal Reserve holdings. As we noted in the issuance section, the Treasury has been lengthening the average maturity of the outstanding Treasury portfolio in recent years and intends to continue this for the foreseeable future. Since these actions by the Fed and the Treasury work in opposite directions it is not intuitive which impact dominates. To discern this we can look at the amount of available securities (the float), within various maturity buckets as a percentage of the total outstanding. To address the impact of the Fed, we use the Fed-adjusted float of Treasuries to make this calculation. In Figure 29 we show the percentage of Treasuries in a variety of maturity buckets at a few key points in time including the present and our forecast for the end of 2014-16.

Figure 29. Aggressive QE Only Maintained Levels on 30yr Treasury Float, This Will Climb as QE Winds Down

| By Maturity Bucket | Pre-Tsy Tsunami 2008-YE | Pre-Op Twist 9/21/2011 | 2012-YE | 2013-YE (f) | 2014-YE (f) | 2015-YE (f) | 2016-YE (f) | New Issuance 2014 |
|---|---|---|---|---|---|---|---|---|
| 0-4yr Maturity | 54.5% | 56.5% | 62.7% | 63.6% | 60.5% | 56.5% | 53.7% | 36.3% |
| 4-6 | 14.5% | 15.7% | 14.9% | 14.1% | 15.6% | 16.9% | 17.1% | 21.8% |
| 6-8 | 8.2% | 8.4% | 8.4% | 8.7% | 8.6% | 8.9% | 9.8% | 18.7% |
| 8-10 | 8.7% | 8.2% | 5.2% | 5.1% | 6.0% | 6.7% | 7.0% | 14.2% |
| 10-20 | 8.8% | 3.1% | 1.9% | 1.4% | 1.2% | 1.0% | 0.9% | - |
| 20-30 | 5.3% | 8.1% | 7.0% | 7.2% | 8.1% | 9.9% | 11.4% | 9.0% |

Source: US Treasury, Federal Reserve and Citi Research

Figure 29 also shows that 15 months of Operation Twist transactions lowered the percentage of Treasury float in the 20yr – 30yr bucket from 8.1% to 7.0%. However, this bucket drifted higher in 2013, despite the significant targeting of long maturities in QE 3. This will begin to rise fairly quickly later in 2014 as QE 3 winds down. Comparing concentrations with prior to the dramatic rise in Treasury issuance in 2009 shows significant rise in 20-year+ maturity Treasuries as a percent of float. Remember that there were only quarterly 30yr bond auctions in 2006 – 2008 and no 30yr bond offerings in 2002 – 2005.

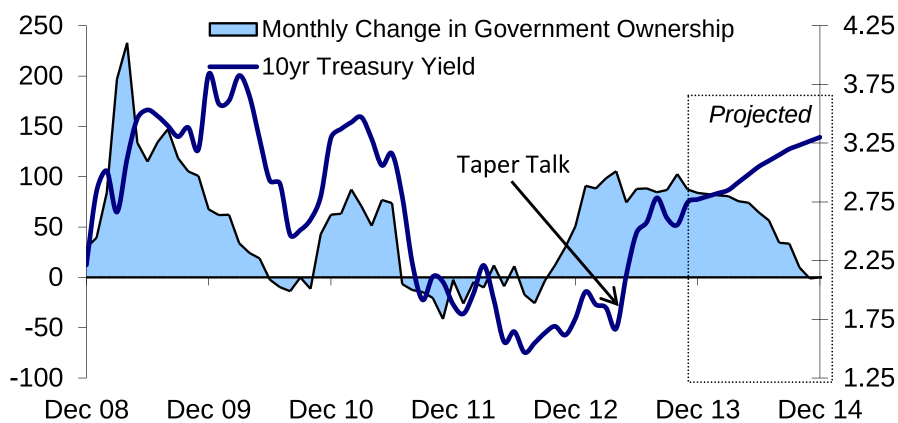
The other notable impact of the Fed programs is that it has significantly increased the concentration of the Treasury float in the very short maturities – the Fed owns very few securities with maturities under 3-years as we noted above in Figure 28.

Note that in 2015 and 2016 the amount of float in the 20-year+ bucket rises well above the percentage of issuance in the 30yr. This is because the float in each bucket is driven not only by issuance, but what rolls out of each bucket, and far more rolls out in the front-end of the curve.

### The Impact of QE Ending on Rate Level May be Limited

The rise in rates mid-year when the tapering of QE was only discussed raise concerns that rates will again rise sharply when QE purchases are actually scaled back. We think that these concerns are misplaced and expect a very modest impact from the ultimate decision to taper. While we do forecast Treasury yields to rise throughout 2014, we expect this to be driven by improving expectations about economic fundamentals more than by the evolving path of QE purchases. While we do not dispute that QE has an impact on Treasury yields, we see this as being in priced in based on anticipation of policy rather than in implementation of policy. In 2013, the May-June period witnessed a dramatic shift in long-term QE expectations and helped drive Treasury yields substantially higher.

Figure 30. Treasury Yields Frequently Move in the Opposite Direction of Fed Purchases




Source: Citi Research

In Figure 30 we show 10yr Treasury yields relative to the change in the size of the Federal Reserve QE portfolio. This shows that yields have often been rising as the Fed was adding to its portfolio and fallen when the Fed has stopped adding to the portfolio. This does not suggest to us that more demand equals lower prices, and vice versa. Instead, we see this as an indication that other factors, such as longer-term fundamental expectations, can dominate shifts in Fed policy.

## Off the Run Treasuries Snapped Up by Fed

One concern that investors have expressed about the end of QE 3 is that it will lead to a significant underperformance of off-the-run Treasuries. However, we think that this is one area where the 'stock' theory of QE will be most helpful. Currently the Federal Reserve owns one-third of all Treasury coupon securities with a maturity of 2yrs or longer. However, it owns over half of those securities that are greater than 5yrs old. In Figure 31 we show that there is a smaller float of 5-year old Treasury coupon securities now than at the end of 2008. When considered as a percentage of total Treasury securities the difference is very dramatic.

Figure 31. Off-the-Run Treasuries Not Very Plentiful

| | Total Outstanding | | | Float, Net of Fed | | |
|---|---|---|---|---|---|---|
| | 2yr+ Mty | 5yrs+ old | % O/S 5yrs+ old | 2yr+ Mty | 5yrs+ old | % Float 5yrs+ old |
| December 2008 | 2,108 | 697 | 33% | 1,842 | 600 | 33% |
| December 2013 | 6,290 | 922 | 15% | 4,227 | 452 | 11% |

Source: US Treasury, Federal Reserve and Citi Research

We would think there is limited room for these securities to perform that poorly given the very limited float.

## The Various Steps of Shifting Fed Policy

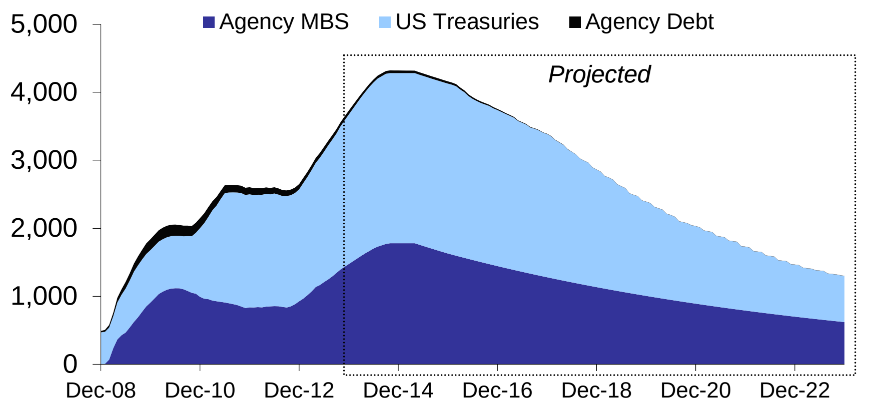
The reduction and ultimate elimination of Fed QE purchases is likely to be the most dramatic shift in Fed policy in 2014. However, there are a significant number of steps that the Fed may ultimately take before it normalizes policy. Keep in mind that the Fed does not consider the reduction in QE purchases to be a tightening of policy, but merely a reduction in the pace of easing of policy. Below we draw upon the Fed's most recent formal discussion of the steps of exit policy, the FOMC minutes from way back in June of 2011. However, we also insert updates to policy that the Fed has discussed since then:

- Gradual taper the amount of QE purchases, currently at $85 billion per month. We expect the initial reduction to be announced at the March 2014 FOMC and the final reduction to be announced at the September 2014 FOMC.
- Continuing to reinvest paydowns for a period of time is likely to be a distinct step, which could last for six or nine months. Under the stock view of QE, this would be considered to be a policy neutral period.
- Ending reinvestment of MBS prepayments and Treasury maturities. This would allow the size of the SOMA portfolio to shrink naturally. While the initial pace of shrinkage would be modest, it would nonetheless have to be considered a modest tightening of policy. This step would likely occur prior to any hike in the Fed Funds rate.

**The growth of the SOMA portfolio in 2014 is likely to be much larger than the pace of shrinkage in future years.**

**We do not expect the SOMA portfolio to return to end of 2013 size until late 2016.**

Figure 32. The SOMA Portfolio May Take Almost a Decade to Normalize




Source: Federal Reserve and Citi Research

- In 2011 the Fed discussed modification of the Fed Funds guidance as a distinct policy step. Since then, the guidance has changed from date-based to data-based which changes this step somewhat. However, there has certainly been no shortage of discussion and interpretation of the current forward guidance. While it is possible that the current guidance could be explicitly altered (i.e. lowering the 6.5% unemployment target) we think that the Fed is more likely to continue to guide the interpretation of this target when it feels market levels are out of line with its own views. This is a less distinct step than those above and it has been used since 2010 and is likely to continue to be used up to, and beyond the eventual initial hike of the Fed Funds target rate.

- Raising the Fed Funds target. We expect the initial hike to come in the second half of 2015. At this step forward guidance may continue to attempt to guide the pace of Fed Funds hikes.

- The Fed is likely to raise the interest rate paid on excess reserves in parallel with the hike of the Fed Funds rate.

- The Fed will use its different repo tools to begin removing liquidity from the system as needed. This is also likely to take place following the initial Fed Funds hike, although it may choose to use the fixed-rate repo tool to raise the effective Fed Funds rate to closer to 25bp prior to the initial target rate hike if this rate remains at the low end of the band approaching the anticipated initial hike date.

- The Fed may potentially choose to sell securities in the SOMA portfolio. In the 2011 discussion of exit policy consensus was that sales would take place sometime after the initial Fed Funds hike and sales in the area of $15-25 billion per year over the course of 3-5 years seemed likely. SOMA asset sales now seem far less likely.

Normalized Fed policy probably consists of a Fed Funds rate that is near 4% and a SOMA portfolio that is somewhere around $1.5 trillion. It is likely to be 2018-19 when the Fed Funds target is reached, and perhaps 2022 before the baseline SOMA portfolio size is reached.[3]

[3] We assume that the baseline SOMA size to be currency in circulation, which is currently $1.2 trillion and we assume grows at 4% per year. This is in line with Fed guidance. We estimate this level would be reached in 2022 given natural runoff. The target could be reached much earlier (and be lower), if the Fed were to sell securities.

# Summary of Fed Extraordinary Actions

## Securities Purchases (QE & Twist)

- Nov. 25 2008 – QE 1
- Mar. 18, 2009 – QE 1 Extended
- Nov. 3, 2010 – QE 2
- Sep. 21, 2011 – Operation Twist
- Jun. 20, 2012 – Operation Twist Extended
- Sep. 13, 2012 – QE 3

Figure 33. Summary of Asset Purchase Programs

| Program | Securities Purchased | Size (in $'s bln) | Average Duration | Program Length (in months) | Total Duration (10yr Equivalents) | 10yr Equivalents Per Month |
|---|---|---|---|---|---|---|
| US Tsy MBS Purch Program (Sep 2008) * | MBS | 240 | 3.5 | 14 | 99 | 7 |
| QE 1 pt. 1 (Nov 2008) | MBS & Debt | 600 | 3.6 | 8 | 253 | 32 |
| QE 1 pt. 2 (Mar 2009) | Tsy, MBS & Debt | 1,150 | 3.6 | 6 | 482 | 80 |
| QE 2 (Nov. 2010) | Tsy | 600 | 5.5 | 8 | 388 | 49 |
| Operation Twist (Sep 2011) | Tsy | 400 | 9.5 | 8 | 447 | 56 |
| Twist Extension (Jun 2012) | Tsy | 267 | 9.2 | 6 | 268 | 45 |
| QE 3 (Sep 2012) – MBS only | MBS | 40/mo | 5.3 | | | 23 |
| QE 3 (Sep 2012) – Treasuries added | Tsy | 45/mo | 9.5 | | | 53 |

Source: Federal Reserve, US Treasury and Citi Research

* Note: Not a Federal Reserve program, but important to consider along with Fed actions

## Extended Guidance

- Sep. 21, 2010 – 'extended period'
- Aug. 9. 2011 – Mid-2013
- Jan. 25, 2012 – Late 2014
- Sep. 13, 2012 – Mid-2015
- Dec. 12, 2013 – 6.5% unemployment, 2% inflation

# TIPS: Hangover

**Jabaz Mathai**
**Kelly Wang**

Real yields are likely to move higher over the course of 2014 as an improving economy continues to reduce the attractiveness of low real yields in favor of riskier assets. Inflation is likely to run low in 2014 and pickup only gradually thereafter. However, breakevens are low relative to our expectations of inflation for 2014 and 2015 and are therefore likely to move higher, especially in the front end. TIPS asset swaps are at attractive levels and offer a good source of return for long term investors.

## Real yields: The selloff should continue in 2014

**Long end real rates sold off significantly this year, and we think the selloff will continue in 2014.**

In our annual outlook for 2013, we had recommended positioning for higher real yields in the long end of the curve. However, back in Dec 2012, we had not anticipated the degree of the selloff that eventually materialized this year, with 10y real yields selling off by 150bp in 2013. Even as the Fed purchased Treasuries every month, and as forward guidance pushed the expected date of Fed funds lift off from the first quarter of 2015 at the beginning of 2013, to the fourth quarter of 2015 as we near the close of the year, longer end real rates sold off substantially and the yield curve steepened.

Real yields will likely be a significant driver of TIPS performance in 2014 and while it remains to be seen whether the change in real yields will be of the same magnitude as in 2013, we think the drivers are the same – an economy that continues to heal at a reasonable rate, and a shift in investor preferences resulting in portfolio allocations away from low real yields and into riskier assets such as equities.

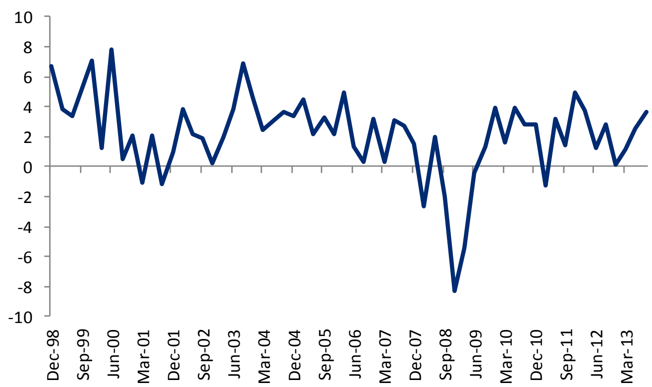
**The economy is likely to continue to improve at a reasonable pace in 2014.**

The economy, which has shown steady improvement over 2013 (see Figure 34) is likely to continue to grow at a reasonable rate in 2014 (our economists forecast 3% real GDP growth in 2014). Fiscal headwinds should moderate in 2014 compared to 2013 as bipartisan budget talks to reduce the impact of the automatic sequester have gained some momentum in late 2013. The debt ceiling should be lifted once again on time in our view, perhaps with some jitters as we get close to the deadline sometime in March, but lifted nevertheless. The labor market should continue to improve - with the unemployment rate dropping to 7% in December 2012 from 7.8% at the beginning of the year, and forecasted by Citi economists to get to 6.4% 2014. The positive momentum in the residential housing market should continue, driven by a combination of low inventory and expectations of continued home price appreciation among investors and homeowners.

**The rise in real rates may be driven more by the belly of the curve than the longer end.**

The rise in real rates may be driven more by the belly of the real yield curve (the 5y to 10y) than the longer end (beyond the 10y). The market might start pricing in a quicker path of Fed funds hikes, especially if the unemployment rate starts falling quicker than market expectations, pushing 5y rates higher. The longer end has largely priced in the effects of tapering and end of QE purchases, as evidenced by the stabilization of long end rates and the recent flattening of the rate curve. The10y20y real rate at 2.15% is likely to trade within a 2% to 2.5% range, as it mostly did in the 2009 to 2011 period. This is also consistent with expectations of steady state real GDP growth of about 2.25% - long end real rates should converge to expectations of long run real GDP.

Figure 34. Real GDP is expected to grow at about 3% in 2014 (Citi economics forecast)



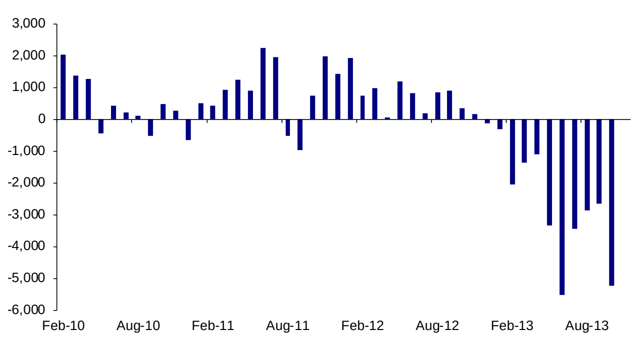

Source: Citi Research

Figure 35. 10y20y real rates are likely to trade in a 2% to 2.5% range in 2014


Source: Citi Research, Bloomberg

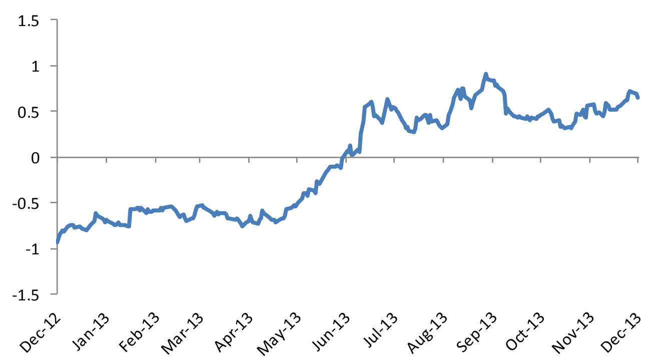
In addition to an improvement in the economic picture, fund flows will continue to have a significant effect on real rates. Figure 36 shows fund flows in inflation linked funds. The rotation out of fixed income and into equities was a key reversal in investor behavior in 2013. Historically, similar changes in investor behavior have tended to persist for long periods – generally well into the hiking cycle. In the current monetary policy environment, investors have reacted to the impending winding down of asset purchases as a de facto tightening, despite the Fed's hope that tapering is perceived as a reduction in stimulus rather than a tightening. Thus, outflows have started earlier than the start of conventional tightening through increases in the fed funds rate, which we expect to be in the third quarter of 2015. This argues for a sustained outflow from fixed income funds through 2014. With inflation not likely to be a problem in 2014 (we examine this in detail in the section on breakevens), investors are likely to continue avoiding inflation linked funds. With breakevens at cheap levels, this supply pressure is likely to be reflected through higher real yields.

Figure 36. Outflows from inflation linked funds are likely to maintain upward pressure on real yields in 2014 ($mm)




Source: Citi Research, EPFR

Figure 37. 10y real rates rise in an environment of fund outflows as QE approaches an end in 2014

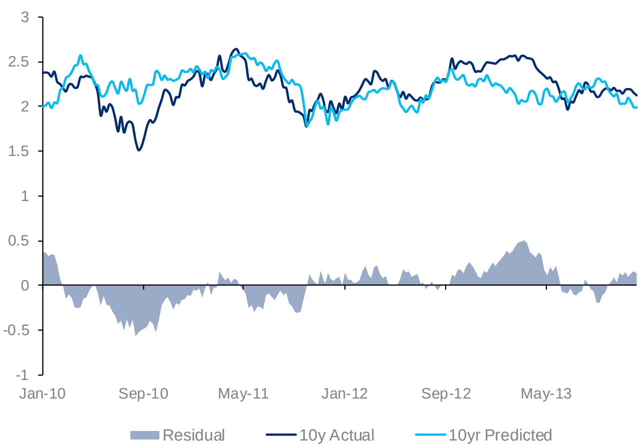



Source: Citi Research, Bloomberg

## Breakevens: Expect low volatility, initial flattening

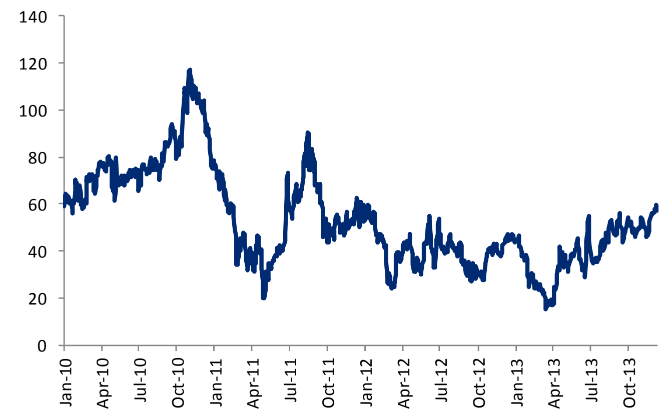
We expect the breakeven curve to move higher from current levels, as the breakeven curve underprices expectations of CPI, especially at the front end. (We expect core and headline CPI to run close to 2% in 2014). The breakeven curve is likely to flatten early in the year, given positive Q1 seasonals and then rise gradually as higher inflation expectations rise slowly. From a fair value standpoint, we see the 10y as slightly overvalued currently based on our FV regression model (see Figure 38). However, inflation expectations are likely to rise during 2014, if our expectations of 3% GDP growth and lower unemployment are borne out. This would provide support for 10y breakevens, which we expect to trade in a range of 2% to 2.5% over the next year.

Figure 38. 10y breakevens are rich compared to our FV model




Source: Citi Research

Figure 39. The 5s30s breakeven curve is likely to flatten initially




Source: Citi Research, Bloomberg

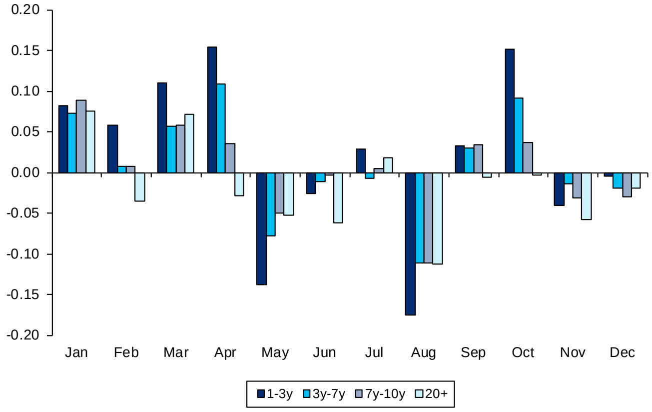
## Position for Q1 front end TIPS outperformance

Front end TIPS have cheapened up due to low realized inflation. One attractive trade to do early in 2014, is buying front end TIPS, especially in the 1y to 2y sector. One year inflation swaps, at 1.5% is roughly at the same level as in Dec 2012. Crude oil however is trading about 11% higher than in mid December ($97.6 vs. $87.7 for the WTI contract, and gasoline is trading at similar levels as last December). Median economist expectations for 2014 and 2015 CPI are 1.9% and 2.1% respectively (source: Bloomberg). We also estimate about 2% for headline CPI next year. If realized inflation is close to median forecasts, then front end TIPS are likely to outperform significantly.

Figure 40 shows the median duration adjusted excess returns of TIPS over nominal Treasuries, by maturity sector, in basis points by month using Citi TIPS index data from 2004 to 2013. It is clear that TIPS have historically outperformed in the first quarter, especially in the 1y to 3y sector (excluding transaction costs). Figure 41 shows the monthly performance data specifically for 2013 – TIPS outperformed in January again this year. Given the attractive valuations in the front end, we expect the front end to do well in early 2014.

The downside to CPI is likely to come primarily from energy, as core CPI is likely to stay firm. This risk can be mitigated by selling out of the money call options on crude oil as a hedge against a long position in front end breakevens.

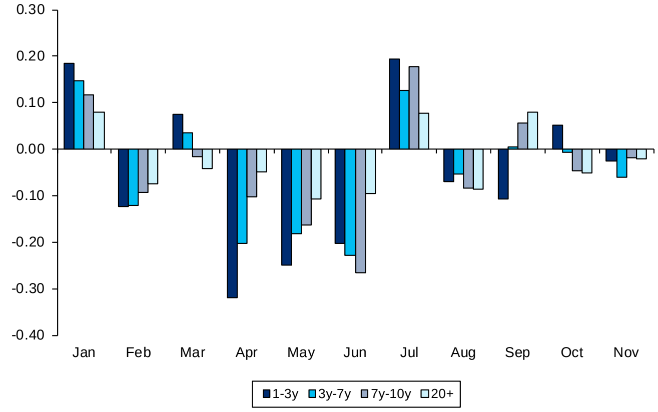
Figure 40. TIPS have historically outperformed nominals in the early part of the year




Source: Citi Research

*Note: Columns represent median bp outperformance of TIPS by maturity sector using TIPS*

Figure 41. In 2013, Q1 was a bright spot for TIPS outperformance



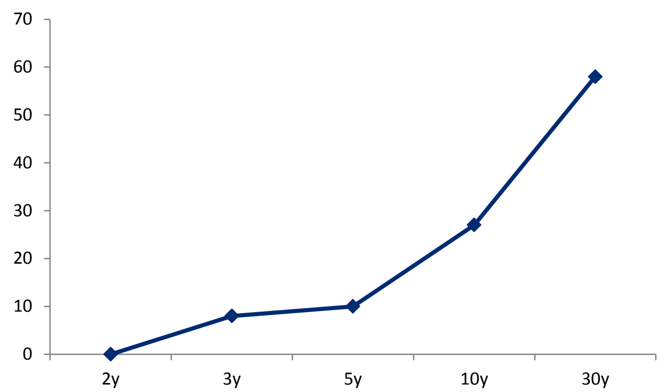

Source: Citi Research

## Opportunities in asset swaps

With the tightening of swap spreads vs. Treasuries, TIPS look attractive again on an asset swap basis. 10y TIPS (the July 23s) can be asset swapped into Libor plus 24bp. This is essentially back to the same levels that existed at the beginning of 2013. For investors looking to generate carry without duration, inflation or credit risk, asset swapping TIPS is an attractive proposition. In addition to earning 24bp over Libor, the rolldown on the asset swap curve is attractive as well at 4bp over the course of a year for the 10y sector.

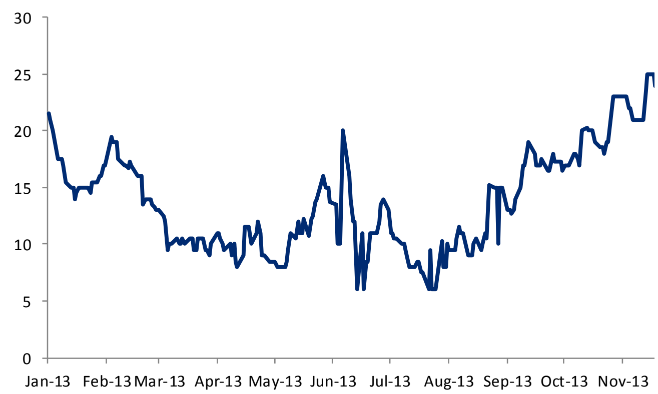
Historically, TIPS have traded cheaper than nominal on asset swap due to lower liquidity. However, we think that any liquidity differential is likely to stay contained, as issuance in TIPS as grown substantially over the last few years. Patient investors are likely to not only earn carry and rolldown, but also get the opportunity to exit at tighter levels sometime in the first half of 2014, in our view.

Figure 42. The TIPS asset swap curve provides good carry for longer term investors




Source: Citi Research, Bloomberg

Figure 43. 10y TIPS asset swap spreads are at attractive levels again




Source: Citi Research, Bloomberg

### 2014 Issuance: Changes in 5y issuance

We expect gross issuance in TIPS to be the same in 2014 as in 2013 at $155bn. However, there are changes planned for the 5y sector. The Treasury, in its November refunding announcement, said that it is studying the idea of offering an additional 5y TIPS cusip. While the details are expected to be announced on February 5 during the first quarterly refunding announcement of 2014, we think that the Treasury would use a new 5y security for October with a reopening of that issue in December. The April issue would likely be opened once. The total number of auctions would edge up to 13 from 12 in our view. The tables below show the 2013 auction schedule and our expectations for 2014.

While we expect gross issuance to be the same, net issuance after redemptions will be a touch lower, with three issues maturing in 2014 vs. two in 2013. Net issuance after redemptions, but not including Fed QE purchases, is expected to be $88bn in 2014 vs. $114bn in 2013. Even after accounting for the end of Fed purchases, we expect net issuance to be lower in 2014.

One interesting characteristic of the market in 2013 has been the relatively good clearing in auctions, despite the underperformance of TIPS. The median tail in 2013 for was only 2.4bp. Four out of the twelve auctions came through the market. The only auction that came poorly was the first 5y auction in April. This was because the increase in size of 5y issuance in 2013 was higher than market expectations (in line with Citi forecasts). Next year, if our expectations of a second cusip in the 5y sector materializes, 5y auctions should go well as the additional auction reduces the issue size per auction in the 5y.

Figure 44. Actual TIPS Supply – 2013 (note, expected for December)

| | 5yr TIPS | 10yr TIPS | 30yr TIPS |
|---|---|---|---|
| 13-Jan | | 15 | |
| 13-Feb | | | 9 |
| 13-Mar | | 13 | |
| 13-Apr | 18 | | |
| 13-May | | 13 | |
| 13-Jun | | | 7 |
| 13-Jul | | 15 | |
| 13-Aug | 16 | | |
| 13-Sep | | 13 | |
| 13-Oct | | | 7 |
| 13-Nov | | 13 | |
| 13-Dec | 16 | | |
| Total | 50 | 82 | 23 |

Source: Citi Research

Figure 45. Expected TIPS Supply - 2014

| | 5yr TIPS | 10yr TIPS | 30yr TIPS |
|---|---|---|---|
| 14-Jan | | 15 | |
| 14-Feb | | | 9 |
| 14-Mar | | 13 | |
| 14-Apr | 13 | | |
| 14-May | | 13 | |
| 14-Jun | | | 7 |
| 14-Jul | | 15 | |
| 14-Aug | 12 | | |
| 14-Sep | | 13 | |
| 14-Oct | 13 | | 7 |
| 14-Nov | | 13 | |
| 14-Dec | 12 | | |
| Total | 50 | 82 | 23 |
| *vs. 2013* | 0 | 0 | 0 |

Source: Citi Research

# Outlook for short-term rates

**Andrew Hollenhorst**

Reserve creation associated with ongoing asset purchases, a flat to slightly negative supply of short-term paper (see our supply projections), and regulation that increases demand for front-end securities while decreasing banks willingness to issue short term liabilities all argue for a continued period of very low short-term rates in 2014.

## Repo rates likely to average between 0-10bp

**Continued reserve creation should keep repo rates at close to 0bp.**

In our view the most important driver of repo rates remains the total supply of Fed reserves. While tapering of Fed asset purchases, which we expect to begin in early 2014, will lead to a slowing in the pace of reserve creation, we project that total reserves will peak at above $3 trillion in late 2014. Before the mid-October front-end rates dislocation associated with the debt ceiling, repo rates had moved down to 5bp or below. While rates have stayed slightly higher over the last month and a half, we think repo rates will move to closer to 5bp over year end. Repo rates may then move higher in late February as seasonal T-bill issuance picks up and the next "hard" debt ceiling date, which we put in March (Figure 47), comes into focus. Following this, we expect repo rates to average in the lower end of the 0-10bp range for the remainder of the year.

**MBS repo rates may stay close to those for Treasuries.**

For much of the last two years, MBS repo has been trading at historically high yields relative to Treasury repo. We have previously attributed this spread to dealer and real estate investment trust (REIT) needs to finance their MBS positions. In 2014 we expect REITs to decrease holdings of MBS implying that the recent MBS/Tsy repo spread compression may be maintained in the coming year.

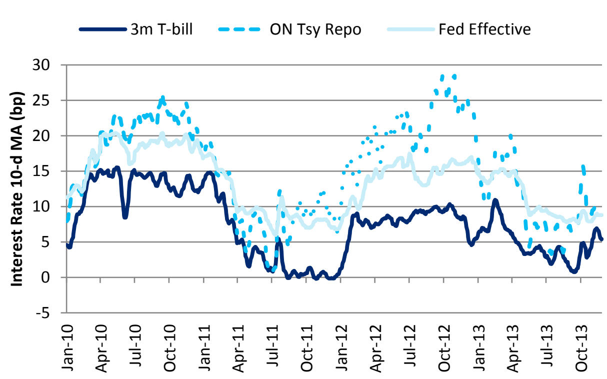
**Expect 6m T-bill rates to decline to 5bp.**

T-bill rates will likely follow a similar path and we think now may be a good time to purchase 6m T-bills which are currently yielding around 10bp, which may come down to 5bp in early 2014. The next opportunity to extend duration may be around the debt ceiling in March. As repo rates drop we think 1m and 3m bills rates will trade at sub-5bp yields.

**Expect Fed to announce another test phase of fixed-rate reverse repo facility.**

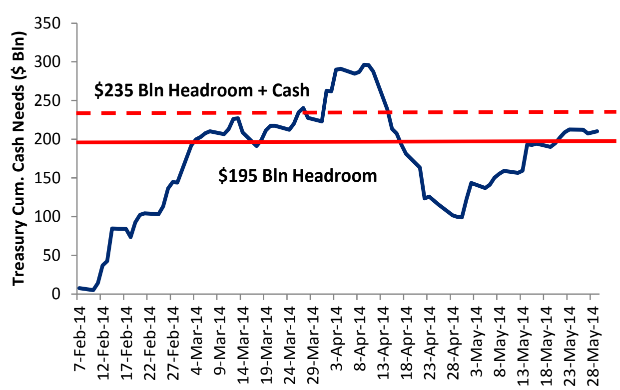
Our view for repo rates could be substantially changed if the Fed were to make an unexpected change to its usage of its new ON fixed-rate reverse repo facility. When the facility is made truly full-allotment it should provide a floor on market repo rates. Currently the facility is in a test phase and given the $1 billion per-counterparty limit and lower than market-rates being offered should not be having a material effect on market repo rates. We think the Fed is likely to want to gain more experience with the facility and may announce another multi-month test phase following the expiration of the current test at the end of January 2014.

Figure 46. Short-term rates may decline toward 0bp in 2014




Source: Citi Research, Bloomberg

Figure 47. Expect next "hard" debt ceiling in March 2014




Source: Citi Research, US Treasury

**Fed unlikely to exert upward pressure on short-term rates until 2015.**

In the new test phase we think the FOMC may give authority to the FRBNY to increase fixed-rate to 10bp or higher (the current limit is 5bp) and possibly increase the per-counterparty limit. In our view, it is unlikely that the Fed will want to use the facility to move rates significantly higher earlier than a few months before it plans to hike rates. If short-term rates were to move negative, which is not our base case for 2014 but certainly a possibility given our low rate expectations, the Fed might choose to make the facility fully operational to avoid disruptions in money markets.

## Expect LIBOR-GC basis to widen

**LIBOR and CP rates may be close to their lows.**

While we expect secured and government short-term borrowing rates to fall further, we think unsecured LIBOR and CP rates may be close to their lows. Even at lower repo rates, we think short-term investors may continue to demand a modest positive yield on short-term CP. With 3m financial CP at around 10bp, we think there is little room for these rates to fall further. 3m LIBOR looks fair relative to CP.

**Cut in IOER would tighten LIBOR-GC, but remains unlikely.**

Speculation increased around a cut in interest on reserves following discussion from a number of Fed speakers and mention in the FOMC minutes. This would upend our rate forecasts for 2014 as it would imply lower LIBOR and higher repo, which would compress the LIBOR-GC basis. However, we think this is not being considered as a near-term option by the Fed and that longer term the Fed would judge the costs associated with a cut in interest on reserves would outweigh any potential benefits.

# Short-term supply outlook

Over the course of 2014 we expect the supply of front-end government securities and repo to shrink by about $65 billion. Reductions in T-bills will be more than offset by issuance of FRNs. Agency discount notes (DNs) will continue to decline and the size of the tri-party repo market may shrink by around 10%.

Figure 48. Supply of front-end government securities may shrink modestly in 2013

| | | 2013 | 2014 (Billion $) | Chg |
|---|---|---|---|---|
| Tsy bills and cpns | | | | |
| | T-bill | 1676 | 1594 | -82 |
| | FRN | 0 | 135 | 135 |
| | Cpn <1y | 1381 | 1395 | 14 |
| Total Tsy | | 3057 | 3124 | 67 |
| Agency Discount Notes | | | | |
| | FHLB | 251 | 273 | 22 |
| | FNMA | 77 | 67 | -10 |
| | FHLMC | 138 | 94 | -44 |
| Total Agency | | 466 | 434 | -32 |
| Tri-party Repo | | | | |
| | Tsy | 615 | 575 | -40 |
| | Agency MBS | 531 | 471 | -60 |
| Total Repo | | 1146 | 1046 | -100 |
| Total | | 4669 | 4604 | -65 |

Source: Citi Research

## T-bill supply will decline to make way for FRNs

Improving deficits in 2013 led Treasury to reduce T-bills outstanding by approximately $80 billion over the year. Citi is forecasting a deficit of $550 billion for fiscal year 2014. Federal government borrowing needs will be closer to $650 billion as in addition to financing the annual deficit, Treasury must also issue debt to finance growth in its portfolio of student loans and other assets. We think calendar year 2014 funding needs will be around $650 billion, relative to a projected $597 billion in Treasury coupon issuance. In addition to coupon issuance we expect $135 billion in Treasury FRN issuance with $15 billion quarterly new issues and $10 billion monthly reopenings.[4] To accommodate the new FRNs, we are forecasting a reduction of $82 billion in T-bills. (Figure 49)

Figure 49. T-bill decline offset by FRNs

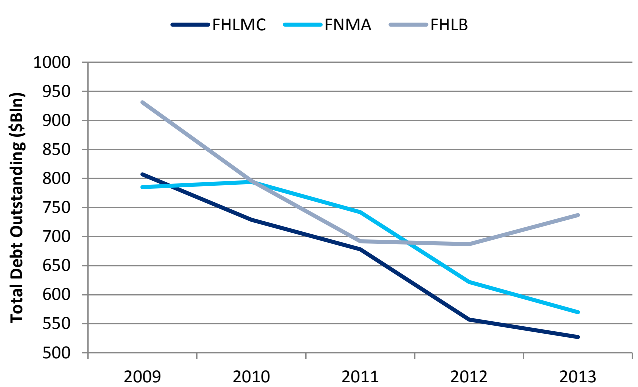
| 2014 Tsy Funding ($ Bln) | |
|---|---|
| Tsy Cpns | 597 |
| FRNs | 135 |
| T-bills | -82 |
| Funding Need | 650 |

Source: Citi Research

## Expect decline in short-term agency issuance to moderate

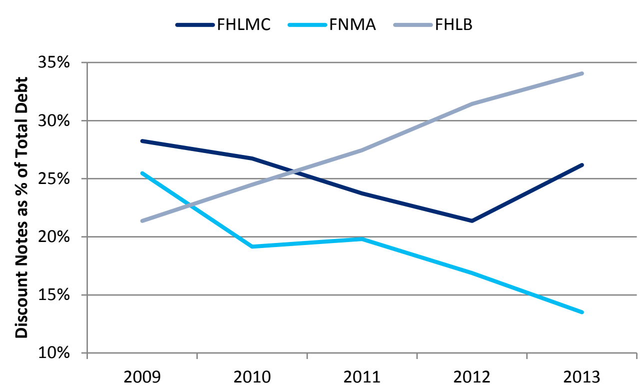
As FNMA and FHLMC wind down their asset portfolios at a mandated rate of at least 15% per year (Figure 50), we expect total debt issuance from these entities to continue to decline. FNMA and FHLMC have also been terming out their liabilities. (Figure 51) While the multi-year trend of less short-term issuance continued for FNMA in 2013, FHLMC actually increased discount notes (DN) as a fraction of total debt. However, the increased FHLMC DN issuance may be partially driven by the need to raise short-term cash ahead of the $30.4 billion dividend FHLMC plans to pay Treasury at the end of December. Longer-term we expect the decline in DNs as a share of total debt to moderate as spread widening for longer term agency debt has made short-term issuance increasingly attractive. (Figure 52) In 2014 we are forecasting that the DN / total debt ratio will revert back to 20% for FHLMC and remain unchanged at 14% for FNMA. Together with our projections for a $158 billion decline in total debt at FNMA and FHLMC, this implies a $10 billion decline in FNMA DNs and $44 billion reduction in FHLMC DNs.

Figure 50. FNMA and FHLMC debt will likely continue to decline…




Source: Citi Research, Debt stats as reported by the agencies

Figure 51. …but the shift away from short-term issuance may moderate

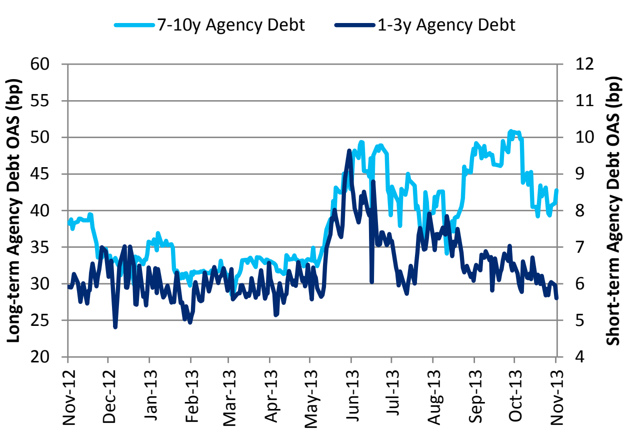



Source: Citi Research, Debt stats as reported by the agencies

[4] We think the first FRN issue auction size may be only $10 billion based on TBAC recommended levels.

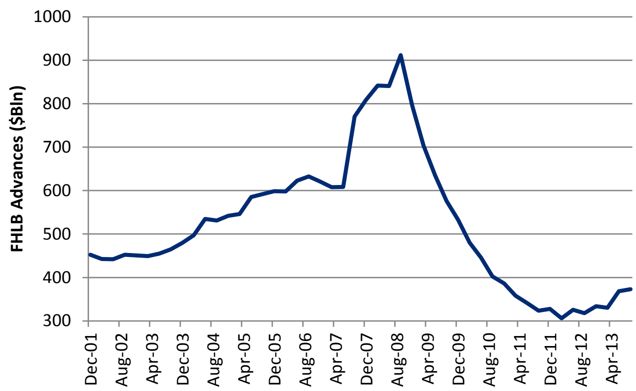
We expect the decline in issuance from FNMA and FHLMC to be partially offset by increased issuance from FHLB. FHLB issues debt to finance the cash advances it makes to member banks. These advances spiked during the crisis period as banks sought to raise liquidity and then quickly fell as deposits poured into the banking system and advances were repaid. (Figure 53) After bottoming in Q1 2012, advances have been rising modestly along with total FHLB debt. While banks in aggregate still have little need to raise additional liquidity, the more generous treatment of FHLB advance liabilities relative to, for instance, deposit liabilities may support the recent trend. As such, we are projecting that total FHLB debt outstanding will increase by $63 billion to $776 billion. Assuming that FHLB slightly increases the share of DNs to 36%, this translates to $22 billion in FHLB DN net issuance.

Figure 52. Tighter spreads make Agency DN issuance attractive




Source: Citi Research, Yield Book

Figure 53. FHLB may need to issue to fund advances

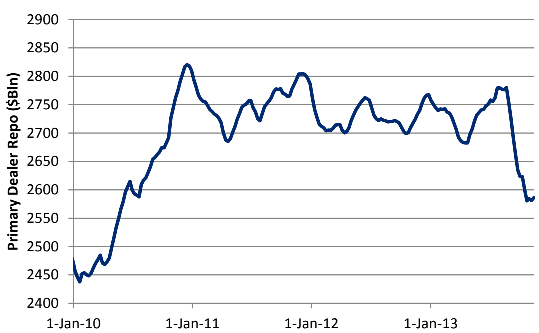



Source: Citi Research, FDIC

## Repo balance sheets may continue to shrink

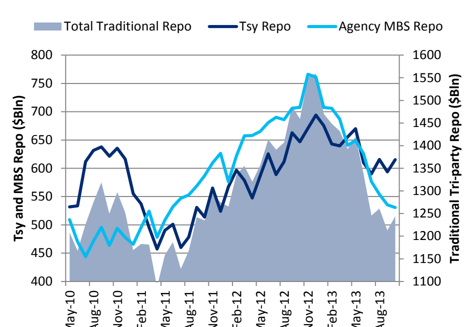
Repo balance sheets at primary dealers have shrunk from $4.6 trillion pre-crisis to $2.6 trillion post-crisis. After remaining fairly stable at between $2.7 - $2.8 trillion, repo balances took a leg down starting this summer with the 6 month moving average now just below $2.6 trillion. (Figure 54) The shrinking size of repo balances has been widely attributed to dealers reducing balance sheet in response to regulation. While regulation will likely be the most important driver of repo balances in the medium-term, we think the recent reduction in repo balance sheets has more to do with levered investors' reduced need to finance MBS collateral.

Figure 54. Repo balances at primary dealers have contracted




Source: Citi Research, Federal Reserve

Figure 55. Decline in MBS collateral driving decrease in tri-party repo




Source: Citi Research, Federal Reserve

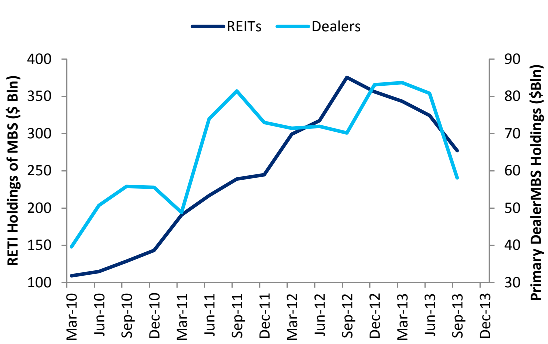
A more relevant measure of available traditional collateral (Tsy and Agency MBS and debentures) is tri-party market repo market size, where short-term investors like MMF invest cash. Data from this market show an approximately $300 billion decline in traditional collateral repo, from about $1.55 trillion in late 2013 to $1.24 trillion as of October. (Figure 55) Over $200 billion of this decline is accounted for by the decline in agency MBS collateral. While in 2011 increasing REIT needs to fund agency MBS in the repo market led agency MBS to overtake Treasuries as the most prevalent tri-party repo collateral, Treasury collateral is now more prevalent than agency MBS. (Figure 55)

The decline in available agency MBS collateral appears to have been driven by levered MBS investors reducing MBS holdings as they became more risk averse around the time tapering of QE purchases was first discussed. Agency mortgage REITs, with a business model based on financing MBS inventory in the repo market, have reduced assets by about $100 billion. Dealers have decreased their MBS inventories by around $30 billion. (Figure 56) Our MBS strategists are forecasting a moderate decline of $10-15 billion in REIT holdings in 2014. Based on this, we think the precipitous decline in repo balance sheets will slow but not reverse.

Longer term, we expect regulation to incentivize the largest banks to continue shrinking repo balance sheets. Citi's bank analysts think the 5% supplementary leverage ratio (SLR) in the US and the 3% Basel III leverage ratio in Europe which will be the binding regulatory constraints in 2014. As we have detailed, the SLR looks at capital / assets with no risk-weighting of assets in the denominator. Since we do not expect banks to raise equity, they will need to cut assets to comply with the SLR. Repo, especially traditional repo, is an obvious choice for cuts as it generates relatively low return and no asset needs to be sold - reverse repo assets and associated repo liabilities can simply be allowed to roll off the balance sheet.

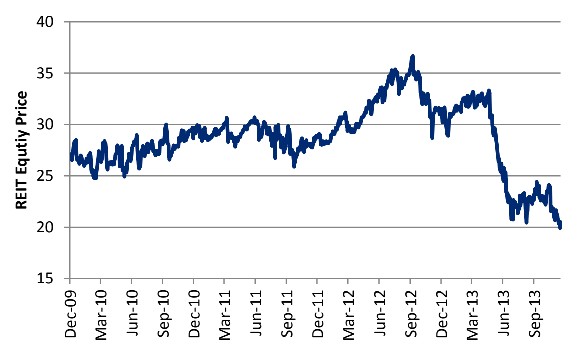
In our view, the extent to which banks will reduce repo balance sheets in response to regulation remains uncertain, especially given considerable unknowns in the final rule. For instance, if the estimated $444 billion in repo assets that sit off-balance sheet under GAAP rules are brought into the SLR denominator, banks would face increased pressure to reduce repo holdings. Even more of a game-changer could be the implementation of a capital surcharge on repo transactions as has been proposed by the Fed. In our view, dealers may not be aggressive in reducing financing they have on with existing clients and will rather discourage adding to repo balances. This implies a somewhat more gradual reduction in repo balance sheet than some in the market are expecting. We think a decrease of $100 billion in traditional tri-party repo might be reasonable and would be surprised to see a decline of more than $250 billion.

Figure 56. REITs reduced MBS holdings as rates rose…




Source: Citi Research, Flow of Funds

Figure 57. …and may not increase assets as equity prices are depressed




Source: Citi Research, Bloomberg, Price for AGNC

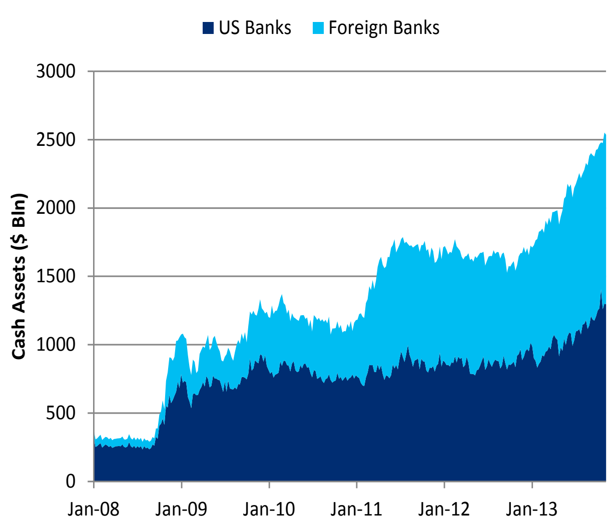
## Financial CP and CDs spurred higher by foreign issuance

As we have discussed elsewhere, foreign banks, not subject to the FDIC assessment, are the most efficient holders of Fed reserves and we think much of the USD CD and CP issuance from these entities has been driven by foreign banks looking to raise cash to place at the Fed to earn 25bp interest on reserves. (Figure 58) Given that we expect Fed reserves to continue to grow in 2014, but at a slightly lower rate than in 2013 as asset purchases are tapered, we forecast a $30 billion increase in foreign bank CP in 2014, less than the $45 billion increase over 2013. (Figure 59) A risk to this forecast is the possibility that the Fed's final rule regulating US offices of foreign banks will be more restrictive than expected, pushing US offices to reduce their short-term borrowing.

Over recent years US banks have been decreasing CP in the market by about $10 billion per year. We think that this trend will continue in 2014. While new regulations like the SLR argue for a faster pace of shrinking balance sheet, recent US short-term debt ratings upgrades argue for increased issuance. We think these effects should partially offset, leading us to project about $80 billion in US bank CP outstanding at the end of 2014.

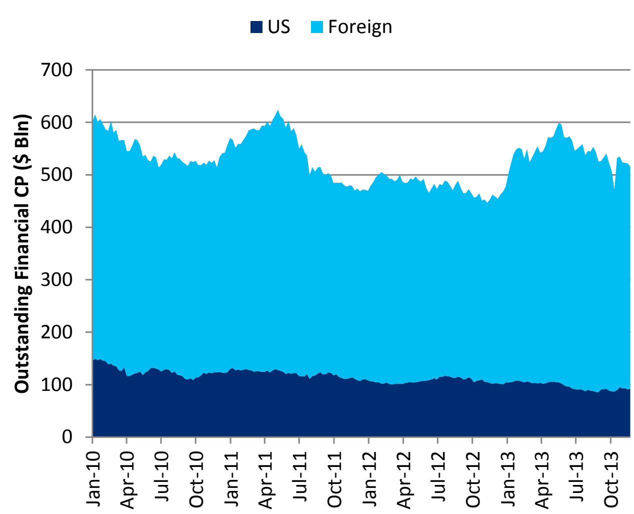
Taking foreign and US bank CP together, this implies modest positive net issuance of $20 billion financial CP, bringing total outstanding financial CP to $538 billion at year end 2014.

Figure 58. Foreign banks holding more cash at the Fed…




Source: Citi Research, Federal Reserve H.8

Figure 59. …has driven issuance of financial CP




Source: Citi Research, Federal Reserve

# Outlook for Interest Rate Volatility

**Neela Gollapudi**
**Anurag Kadasne**

- Our outlook for 2014 calls for subdued realized and implied volatility for the year on account of a) a perception and possible reality of reduced economic uncertainty, b) a starting level of rates that are significantly closer to fair value, and c) a historical precedent that post crisis economies experience materially lower economic volatility.
- We estimate average realized volatility in 10y rates in the low to mid 60s (bp/annum). We expect 1m10y and 1y10y implied volatility to trade in a 65-75bp/annum and 70-85bp/annum through much of the year.
- We expect relative valuation of gamma on 5y vs gamma on 10y rates to continue trading directionally with the level of rates.
- Long-expiry vol appears cheap, both on an outright basis, as well as relative to where rates are. While such vol could drop further with the overall level of vol, as well with higher rates next year, we expect such declines to be modest.
- High strike skew in 2y2y offers the most potent hedging vehicle against higher rates, on a delta-hedged basis.
- We find high strike skew in 10y tails to be modestly rich, while high-strike skew in 30y tails to be either fair, or possibly even cheap. 6m30y payer 1by2s are an interesting way of expressing modestly rate bearish views.
- Finally, we find low strike vol in long-expiry long-tenor swaptions an attractive buy to position for non-consensus rate outcomes of lower long forwards.

There are three factors that drive our core outlook for interest rate volatility – economic volatility, potential realized volatility of rates given economic volatility, and risk premium the market would charge given some level of realized volatility.

## Economic volatility

We measure economic uncertainty in three different ways. We look at the range of economists' forecasts for one-year-ahead core PCE, output gap[5] and GDP growth, both at the end of last year, as well as at the present time. From these forecasts, we estimate a range of forecasted fair values for 10y rates one year out – both at the end of last year as well as this year. This gives us a quantitative sense as to how perceived economic uncertainty evolved during the year. We conduct a similar exercise using the Fed's forecasts for economic activity.

In addition we also look at the time series variability of "10y fair value" over the year 2013, to make assessments of what economic volatility had been during the year. If the near-past were representative of the near-future, then we would know what to expect for 2014.

Figure 60. Private economists' forecast variability has declined since last year, but Fed forecast variability has increased

| | | Bloomberg Economists' 2012 forecasts for 2013 | Bloomberg Economists' 2013 forecasts for 2014 | | Fed Reserve Board Members 2012 forecasts for 2013 | Fed Reserve Board Members 2013 forecasts for 2014 | | 2013 Realizations |
|---|---|---|---|---|---|---|---|---|
| Core PCE | Average | 1.71 | 1.63 | Average | 1.75 | 1.70 | Average | 1.36 |
| | Std Dev | 0.23 | 0.19 | Range | 0.50 | 0.60 | Std Dev | 0.23 |

[5] Output gap is defined as the difference between NAIRU and the unemployment rate. NAIRU is the non-accelerating inflation rate of unemployment that is published by the OECD among others.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Real GDP | Average | 2.03 | 2.61 | Average | 2.60 | 2.75 | Average | 1.83* |
| | Std Dev | 0.30 | 0.30 | Range | 1.20 | 1.10 | Std Dev | 1.54* |
| Output Gap | Average | 1.62 | 0.79 | Average | 1.27 | 0.47 | Average | 1.46 |
| | Std Dev | 0.21 | 0.18 | Range | 0.90 | 0.70 | Std Dev | 0.27 |
| Imputed 10y rate | Average | 4.05 | 4.32 | Average | 4.33 | 4.57 | Average | 3.55 |
| | Std Dev | 0.40 | 0.31 | Range | 0.61 | 0.81 | Std Dev | 0.13 |

* Quarterly data

Source: Citi Research, Bloomberg, Federal Reserve; The last column showing 2013 realizations shows the average value and standard deviation of the actual economic values being forecast, as well as that of imputed 10y rates. The standard deviation for 10y rates uses levels instead of changes.

The variation in the estimates of inflation, output gap and GDP feed into the variability of the imputed 10y rate. A few interesting observations are apparent from the data:

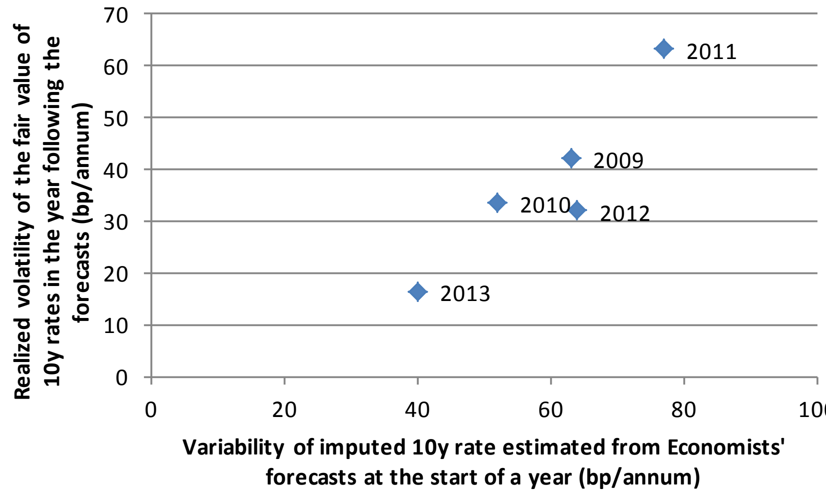
- Private sector estimates of economic variables driving the variability of the 10y rate have declined from 2012 to the present time. In fact, we will find below that the variability in private sector forecasts has declined steadily over the past 6 years as realized economic volatility has remained subdued. The variability of the imputed 10y rate from Fed forecasts has increased over the same period.
- Even though, optically, there is uncertainty in all three drivers of rates, most of the variability in the imputed 10y rate is coming from the uncertainty in inflation forecasts due to the large sensitivity of the 10y rate to core PCE.
- The volatility of the "fair value" of 10y rates over the past year has been only 13 basis points, in sharp contrast to the extremely large realized volatility in nominal 10y yields. This suggests that economic volatility is only one component of realized volatility of 10y rates.
- Based on how sensitive rates are to the three drivers we consider here, surprises in inflation have the largest scope for creating volatility in the fair value of 10y rates. Given that current core PCE YoY is at 1.1, consensus estimates of core PCE appear far away from where inflation is currently printing. If core PCE prints 1.1 (as it currently does) vs median forecasts of 1.63, the difference is worth about 75bp on the fair value model for the 10y rate.

## Why do we care about economists' forecasts?

A prudent reader might ask why we should even look at economists' forecasts. We offer a three part argument.

- We do not have anything better to ascertain economic uncertainty, apart from looking at the vol markets.
- Ergodic theory offers some support for the idea of a linkage between cross-sectional measures of uncertainty and time series measures of uncertainty. It should be noted that we are obfuscating the difference between an individual forecaster's cross-sectional sense of uncertainty and the range in a group of forecasters' point estimates for the central tendency of a variable.
- Most importantly, we can offer some empirical evidence that the variability in the imputed 10y rate arising from the variability in the economists' forecasts of core PCE, Output gap, and GDP appears to be somewhat related to the actual time series variability of the fair value of the 10y rate through the following year.

Figure 61. The variability of economists' forecasts appears to have some predictive power for variability of economic data in the following year




Source: Citi Research, Bloomberg

## Noise volatility

Over the past year, many market participants' estimate of fair value for 10y rates has hovered between the high 2% and the low 3% area[6]. Actual 10y rates have started the year in the mid 1% area. Naturally, adjustment is expected, as we have seen through much of the year. This has nothing to do with estimates of business cycle uncertainty. Instead, it is an unwinding of the Treasury-bid from 2011. Having unwound most of the flight-to-quality premium, Treasury yields may be expected to experience only a modest level of "noise", other things equal, during 2014. Looking at standard deviation of the residuals between actual 10y yields and the fair value of 10y yields since 1987 suggests a "noise" volatility of about 22bp.

Market noise in 2014 will likely be substantially lower than market noise in 2013, simple because rates are starting much closer to fair value. What is different in 2014 though are two new issues – a) a modest decline in Fed credibility arising from perceived poor communication surrounding the September taper non-decision, and b) a decline in market liquidity due to impaired risk capacity at both dealers and other market participants.

Noise volatility in 2013 has been as high 40bp/annum. This should place some bounds on what is plausible due to Fed credibility type reasons. A large part of the 40bp/annum we think can be attributed to the initial low starting point of rates. We would be comfortable ascribing no more than 10bp/annum to Fed-credibility type concerns.

The impaired risk capacity that we mention above pertains to required higher capital requirements at the dealer, as well as higher margins at the clearing houses for derivative transactions. While reduction of risk capacity at the dealers likely increases short-term volatility due to reduced intermediation / buffering, it is less clear how reduction in risk taking by other market participants due to higher margins affects volatility. Reasonable arguments could be made that less risk-taking overall

[6] This estimate assumes a 100bp of term-premium reduction due to Fed QE.

could even reduce realized volatility. We do not think impaired dealer risk capacity should affect anything beyond 1m expiries of realized volatility.

## Piecing together realized volatility

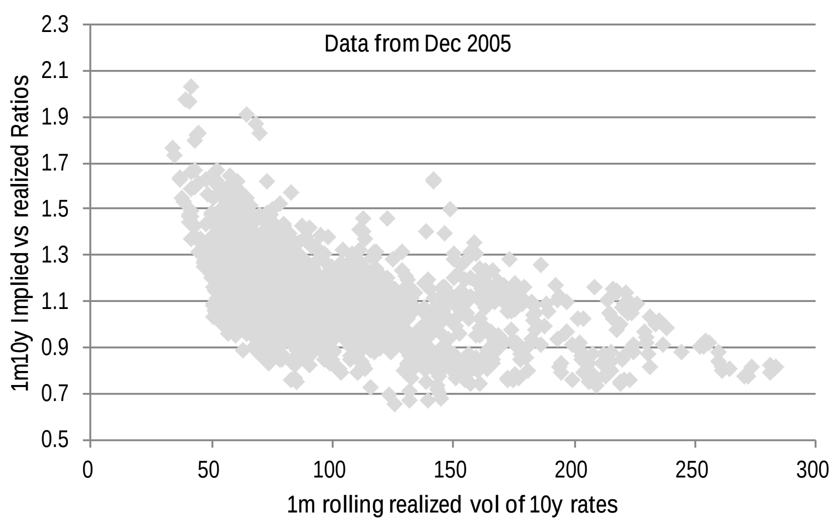
Conservatively assuming that there is 100% correlation between economic volatility and noise volatility, the standard deviations are additive. Adding 30bp/annum of economically driven volatility (this is the standard deviation in the imputed 10y rates coming out of economists' forecasts for core PCE, Output Gap and GDP), 22bp/annum of long-term noise (26 years of annualized standard deviation of the difference between actual 10y rates and the fair value of 10y rates), and another 10bp/annum of Fed-credibility driven noise should offer us ambient realized volatility of about 62bp/annum for the year. The "quite" 2012 saw 72bp/annum, while the *busy* 2013 saw only about 78bp/annum of realized volatility.

The 30bp/annum of economic volatility is likely an overestimate, as the economists' forecast variability traditionally overestimated actual realized business cycle volatility. Also, a possible miss on economists' estimates of inflation (core PCE is printing far lower than estimates) would be a vol depressant, as the market would likely be reluctant to rally in the face of a cyclical growth upswing. In short, we think the above "low 60bp/annum" realized volatility estimate for the year has downside risks.

## Pricing Gamma

Given realized volatility of 62bp/annum, average implied vs. realized ratios are around 1.25. We suspect that actual implied vs realized rations will trade closer to 1 or 1.1. This would place the typical level of 1m10y vol for the year somewhere between 65 and 75bp/annum. We would not be surprised if it dips into the 50s.

Figure 62. 1m10y vol should trade between 65 and 75bp/annum given a forecast realized vol of the 10y rate at 62bp/annum during the coming year

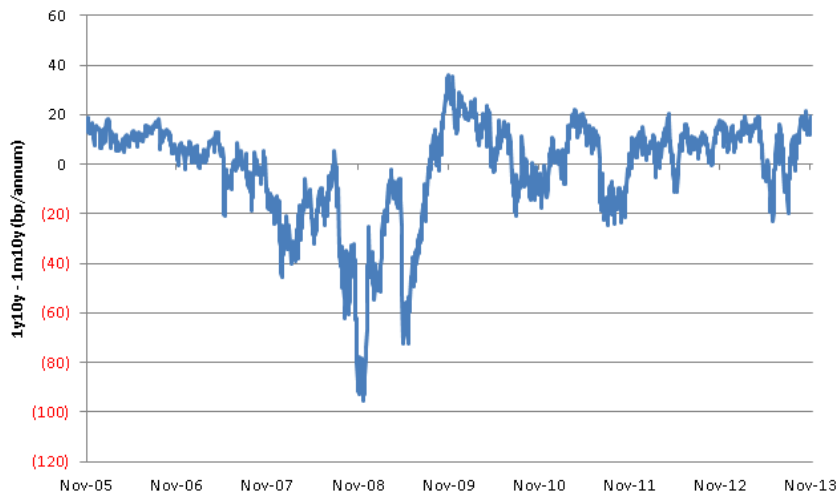



Source: Citi Research

## Pricing Intermediate Volatility

The slope between 1m10y and 1y10y vol reached as high as 35bp/annum, and as low as -95bp/annum during the peak of the crisis over the past seven years. More realistically, during 2006 it averaged 11bp/annum, and during 2012 it averaged 7.5bp/annum. These are templates of what we see as plausible for 2014.

Figure 63. We see the 1y10y vs 1m10y vol basis to look like the experience from 2006 or the experience from 2012



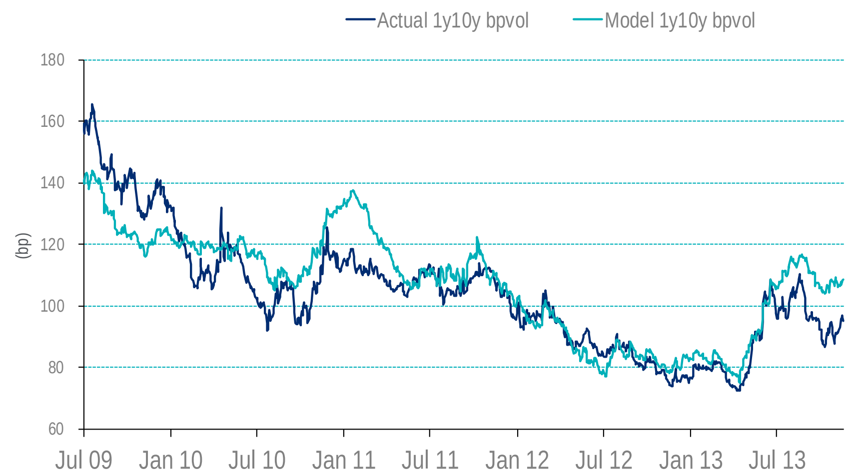

Source: Citi Research

The reason why we see 2006 and 2012 as templates is the very low level of realized volatility. These periods represent both very low and high rates, as well as steep and flat interest rate curves.

Under normal conditions, the slope of the curve is jointly determined together with the level of interest rate volatility. Various market participants believe that with the yield curve as steep as it currently is, intermediate volatility should be higher. While we generally agree that vol and the curve have been historically related, we think this link is not properly understood. We think the link exists as both the curve and vol were traditionally driven by Fed activity.

Given that the very front of the yield curve is very flat over the next year, reflecting subdued Fed activity, 1y10y implied volatility should be very subdued as well. In this context, 1y10y implied volatility should not be driven by the shape of the entire curve from 2y to 10y, but only by the front-end from the 3m bill yield to the 1y or 2y point. The portion of the yield curve that is out beyond the 5y point should not normally affect where 1y10y should trade. Therefore we expect vol to look cheap versus a historically calibrated vol model that uses the slope of the 2s10s curve as one of its drivers.

Figure 64. 1y10y vol looks cheap to model; It should cheapen further next year vs. model




Source: Citi Research; Model 1y10y = 25.5 + 13.9 * (2s10s slope) + 0.37 * (10y 3m RV) + 6.7 * (10y Rate)

Given a slope of the vol surface between 1m10y and 1y10y of 5 to 10bp/annum, 1y10y fair value could range between 70 and 85bp/annum – both levels being below current levels of 1y10y. This forecast for 1y10y should be seen in the context of 10y rates ending in the low 3% area at the end of 2014, and reflect the view that 1y10y could trade significantly cheap to a historically calibrated model (see Figure above).

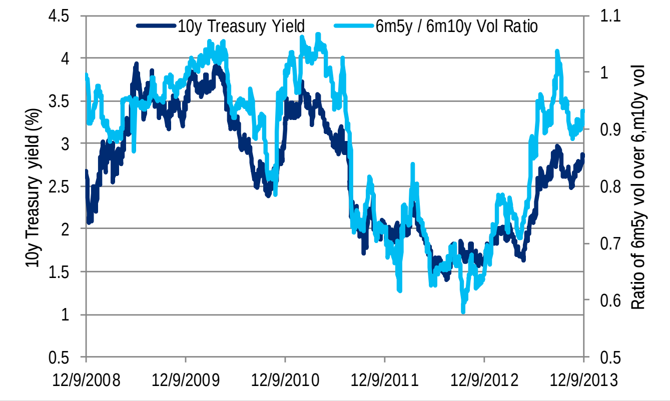
Notwithstanding the slope of the front-of-the-curve vs slope of the back-of-the-curve argument vol is still about 5bp/annum cheap to the 3mL vs ED4 curve (vs the 10-15bp/annum of cheapness against the 2s10s curve). The “cheapness” of intermediate vol when compared to rates market variables suggests that carry earned through steepeners could be of better quality than carry earned through selling vol, especially on a delta-hedged basis. This has in fact been the case recently.

The only way this phenomenon can be rationalized is by understanding that the marginal seller of vol has likely become progressively un-hedged (as opposed to being delta-hedged), and in a mean-reverting environment, leaving positions un-hedged can allow an investor to experience effectively lower realized volatility against short gamma positions. Just to reiterate our point from above, we expect this cheapening to continue further.

## Gamma RV on the curve

We comment on the valuation of 6m5y vs. 6m10y.  We find that the ratio of 6m10y vs 6m5y is strongly directional with the level of rates. The directionality itself has been declining though, after having peaked in May.

Figure 65. 6m5y vol as a ratio of 6m10y vol is strongly directional with the level of rates




Source: Citi Research

Figure 66. The degree of rate directionality of the vol ratio 6m5y vs 6m10y has declined from May, as rates rose, and subsequently when tightening and tapering became delinked.

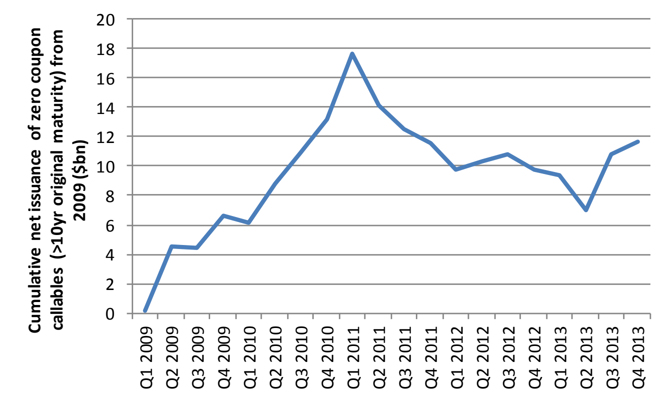

Source: Citi Research

We think the directionality between the 6m5y vs 6m10y vol ratio and rates will continue. We don't think the weakening in directionality will persist though, as further material increases in rates will likely call into question whether the Fed can truly stay on hold in the face of improving data.

## Long-expiry vol

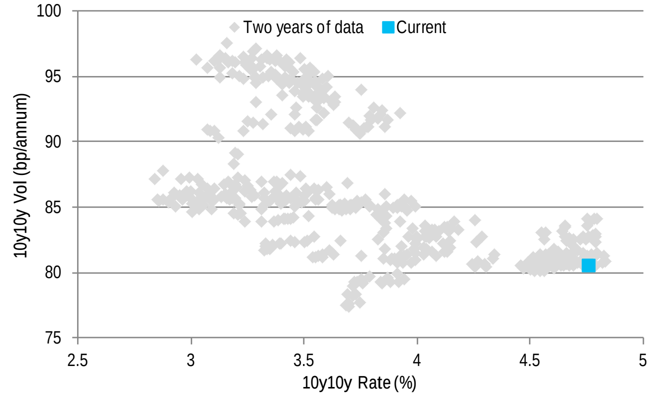
Long expiry vol fell as rates rose May through August, but did not rise as rates retraced part of the rate move after September of 2013. The reason for this is a pickup of issuance of zero coupon callables starting the third quarter of 2013 given elevated rates.

Figure 67. The market saw positive net issuance of zero coupon callables for the first time in over two years




Source: Citi Research

Figure 68. 10y10y Vol is at low-end of range, both outright, and given level of rates




Source: Citi Research

The market saw positive net issuance of zero coupon callables for the first time in over two years, after the GSEs stopped issuing them, followed by issues being called at much lower rates in late 2011 and early 2012.

We think the market for zero-coupon callables is limited to legacy investors that have developed some comfort with the product, and recent new issuance is mostly investors re-investing proceeds from called paper from several quarters ago. Therefore, it is not clear to us that we will see the large seasonal first quarter gross issuance in early 2014. That flow may have happened one quarter earlier this time around. Therefore there is a smaller likelihood of the usual large, rate-insensitive declines in long-dated vol in January and February.

As we mention above, we find the level of 10y10y vol cheap, both on an outright basis, as well as in relation to the level of rates. Still, given our general bearish view on vol, and the modest bearish view on rates (vs spot; not vs. forwards), we would not be buyers of long-expiry vol outright at current levels.

On the other hand, given some cheapness in low-strike skew (that we discuss below), we like buying low strike long-expiry vol in receiver form, as an outright rate trade that carries well.

## Upper-left skew

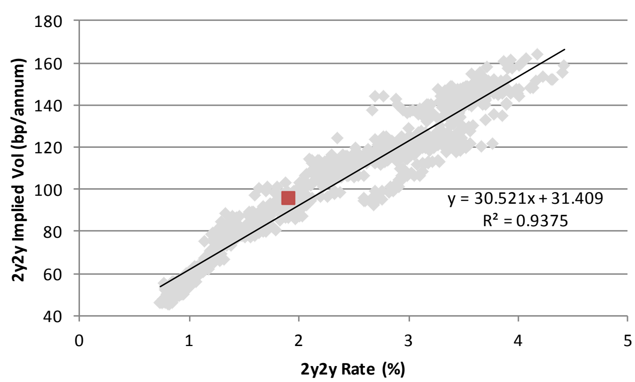
Upper left high-strike skew, specifically in 2y to 3y expiries on 1y to 2y tenors, has performed extremely well in rate backups on a delta-hedged basis. The performance has been strong enough that it could even offset the benign negative carry in times when rates are declining, or are range-bound.

Figure 69. Average daily delta-hedged returns to holding atmf+50bp high strike payers from the start of 2010

| | 1y | 2y | 5y | 10y | 30y |
|---|---|---|---|---|---|
| 1m | -7.1% | -7.3% | -3.2% | -2.5% | -0.1% |
| 3m | -1.0% | -0.8% | 0.1% | -0.2% | 0.1% |
| 6m | -0.26% | -0.10% | 0.15% | -0.01% | 0.03% |
| 1y | -0.08% | 0.09% | 0.15% | 0.06% | 0.03% |
| 2y | 0.11% | 0.14% | 0.13% | 0.08% | 0.05% |
| 3y | 0.14% | 0.14% | 0.11% | 0.08% | 0.05% |
| 5y | 0.11% | 0.10% | 0.08% | 0.07% | 0.05% |

Source: Citi Research

Figure 70. 2y2y Swaption vol is strongly directional with rates; Realized skew in this sector suggests 2y2y low strike skew is rich




Source: Citi Research

As we can see from the table on the left, the 2y2y part of the surface returned a daily average of 0.14% (of inception premium) on long positions in atmf+50 high strike payers over the entire period from 2010, including periods of selloffs, rallies and side-ways markets. Most of these returns come from times when rates backup sharply, but these structures carry very benignly at other times. Therefore, we think delta-hedged upper-left high-strike vol offers the best choice to hedge against higher rates.

Looking at low strikes, we note that atmf-50bp receivers in 2y2y are priced 10bp/annum below atmf vol. On the other hand, looking at the chart on the right above, realized skew runs about 30bp/annum for a 100bp move in rates, or about 15bp/annum per 50bp move in rates. This suggests that low strike skew in 2y2y vol is modestly rich. In fact even atmf vol looks modestly rich looking at the above chart, given where rates are.

One way to express the view that both atmf vol and low strike skew appear modestly rich is to construct 1by2 type structures using receiver swaptions.

## High strike skew in 10y and 30y tails

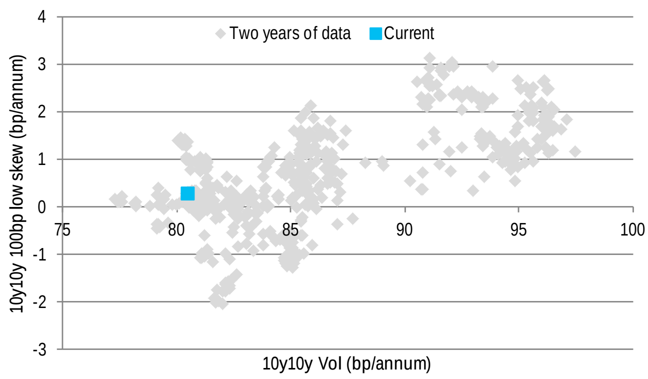
The backup this year in rates ran from the start of May to the end of August. Over this period, 1y10y swaption vol rose 72bp/annum to 110bp/annum. Over this period, 1y10y forwards rose by 150bp. The "realized" skew is about 25bp/annum for a 100bp move in the forwards. At the present time, 100 high skew in 1y10y is priced at 13bp/annum. This suggests that the market is pricing in a greater than 50% chance of rates moving higher than forwards. We disagree with the idea that the odds of 10y rates beating forwards, on the way higher, are greater than 50%. If anything, we could see rates being either unchanged, or even move lower on a taper announcement by the Fed. Therefore, we see high strike skew in 10y rates as being modestly rich.

A comparable analysis in 30y tails suggests current pricing at +2.5bp/annum for atmf+100bp high skew. Over the same May through August period this year, 1y30y atmf rose by 21bp/annum over a 121bp move in rates. This still suggests a positive realized skew of 17.8bp/annum per 100bp move in rates. Therefore, given pricing of only +2.5bp/annum for the 100 high skew, we think valuations are either optically "fair" or even "cheap". What the market probably has priced in though, is the possibility that further backups in 30y rates will be far more reluctant, and vol may not be as directional with rates in such scenarios. We may even be tempted to consider 6m30y 1by2s in payers as a rate trade to express the view that realized vol, especially in the 30y sector, will be subdued in further rate backups.

## Low strikes in long expiry options

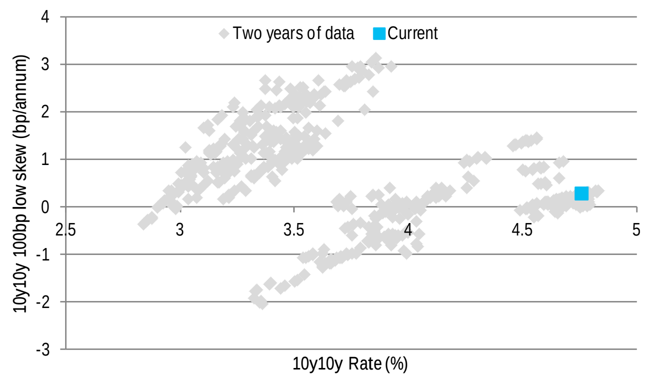
We find low strike skew in 5y10y, 10y10y and 2y30y vol modestly cheap on a historical basis given the level of rates, although skew appears fair purely in relation to the level of vol.

Figure 71. 100bp low strike skew in 10y10y looks fair given level of vol




Source: Citi Research

Figure 72. .. but looks modestly cheap given level of rates; The "cheapness" is perhaps not economically significant




Source: Citi Research

We would have expected a somewhat larger demand for low strike skew given how steep the curve is, and the fact that long forwards are within 50bp of post-crisis highs, and indeed within 50bp of levels seen just before the crisis in 2008. At around these levels, and at horizons greater than 2y, we find the risks to rates somewhat more symmetric than what is indicated by skew market pricing, and we are surprised why we have not seen more flow from end users of long expiry vol. We would buyers of low strike long expiry options on long rates on a rate basis, as we discussed above.

# Outlook for Swap Spreads

**Andrew Hollenhorst**
**Kelly Wang**

In 2014 we expect swap spreads to widen. Front-end spreads are tight relative to the widening we expect in the LIBOR-GC basis over the remainder of 2013 and early in 2014. Further out the curve, 10y spreads are tight relative to fair value based on government deficits. While some commenters view regulatory developments, particularly the SLR, as a significant reason for spreads to tighten in 2014, we see this as a less important and straightforward driver of spreads.

## Front-end spreads may widen with LIBOR-GC

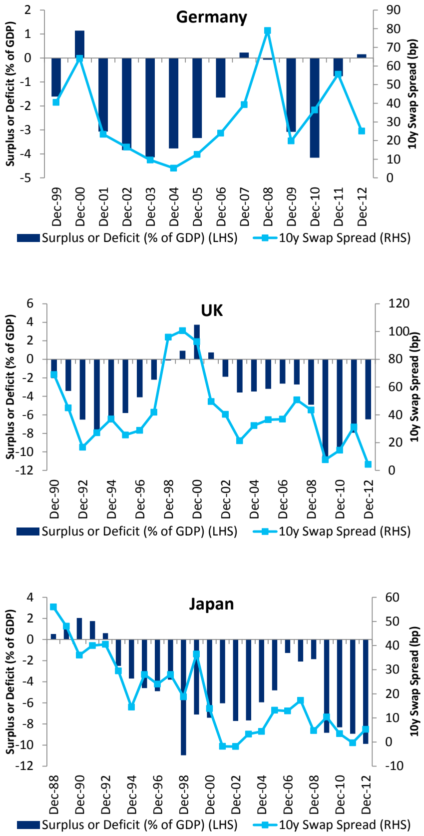
As we previously laid out, we expect the LIBOR-GC basis to be the most important driver of front-end swap spreads in 2014. Since bottoming at just above 23.5bp in late October, 3m LIBOR has moved sideways, staying around the 24bp level. (Figure 74) With 3m LIBOR seemingly at its floor, repo rates will become the more important determinant of the basis. Given our expectation for continued reserve creation coupled with declining investable short-term securities through much of 2014, we expect repo rates to head lower through year end and into 2014. We think the LIBOR-GC basis may widen by about 5bp. The historical beta between 2y swap spreads and LIBOR-GC implies 2y spreads should then be about 3-5bp wider or around 14bp.

## 10y swap spreads are tight to fair-value based on deficits

We see even more room for potential widening in longer term swap spreads. Previously, we have pegged fair value for 10y swap spreads at around 16bp, about 9bp wider than current levels. We arrive at this fair-value primarily by relying on the historical relationship between deficits and swap spreads, the idea being that smaller deficits means scarcer supply of cash Treasuries and hence wider swap spreads. This relationship has been very reliable both across time (Figure 75) and across various countries (Figure 73). We acknowledge that tapering of QE purchases should mean more cash Treasuries in the market and hence tighter swap spreads. However, in regressions either current QE purchases or expectations of future QE purchases have little explanatory power. We have attempted to roughly account for the impact of tapering by adjusting 10y fair-value tighter by looking at how tight spreads have traded since tapering was discussed in May. The 16bp fair value at the 10y point incorporates this adjustment.

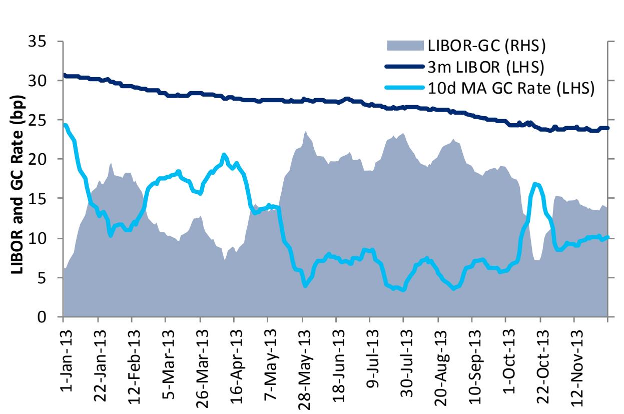
Overall, we think that swap spreads will widen in 2014 with 2y spreads ending the year around 14bp, 10y 16bp and 30y -5bp. This implies that the swap spread curve will steepen by 4bp between 2y and 10y and flatten by 6bp between 10y and 30y.

Figure 73. Swap spreads tracked deficits well globally



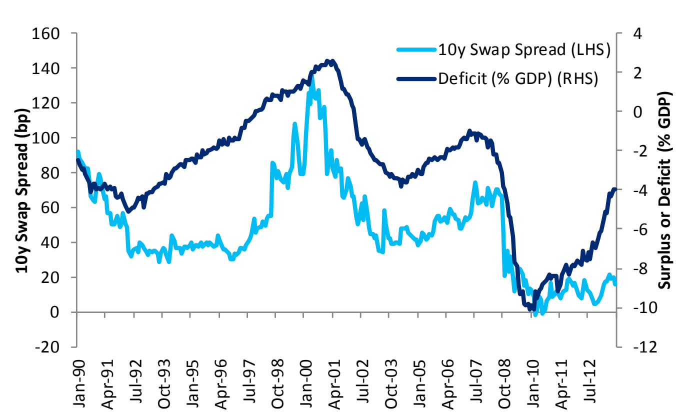

Source: Citi Research, Bloomberg

Figure 74. LIBOR at its floor, repo rates drive front-end spreads




Source: Citi Research, Bloomberg

Figure 75. 10y Swap spread has tracked deficits well across time




Source: Citi Research, Bloomberg

## Basel III SLR may not be a significant spread tightener

**SLR unlikely a main driver of swap spreads in 2014.**

While we agree that the SLR is likely to lead banks and dealers to hold smaller Treasury inventories and reduce the size of repo books, we do not agree with those who go on to conclude that this means demand for cash product will substantially decrease and swap spreads will tighten. Banks currently hold Treasuries to comply with other regulations (e.g. LCR) and we do not think the SLR will cause them to substantially reduce Treasury holdings. Dealers hold around $60 billion in Treasury coupons and while SLR may cause this to decline, at less than 1% of total outstanding $60 billion is far too little, in our view, to be a driver of spreads.

**A dramatic reduction in REIT assets is unlikely in 2014.**

**Levered investors are less likely to have long-only cash Treasury positions.**

This leaves levered investors who may find it more difficult to finance cash Treasuries in repo markets. For instance, REITs hold around $300 billion cash MBS and as repo funding becomes scarcer due to the SLR, some have speculated they may look to sell securities. According to this view, MBS sales would cheapen other cash duration assets like Treasuries, causing swap spreads to tighten. We think that such a dramatic reduction in REIT assets is unlikely in 2014 and even if it were to occur it would be much more a story for MBS spreads than swap spreads. Other levered investors are less likely to have long-only cash Treasury positions. Finally, we think projected reductions in repo balance sheets may be overblown and we are projecting that tri-party Treasury repo may shrink by only about $40 billion in 2014. While this does not take into account bilateral repos, it still gives a sense of the relatively small magnitude reduction in Treasury demand we expect in 2014.

**Other regulations support wider swap spreads.**

Outside of SLR, other regulation argues for swap spread widening in our view. For instance, increased demand for collateral to post against derivative and mortgage TBA transactions should increase demand for front-end treasuries, widening front-end swap spreads. LCR, which explicitly incentivizes banks to hold Treasuries should also be a swap spread widener. Taking the doubts we have about the quantitative impact of the SLR on swap spreads together with countervailing effects from other regulation, we maintain our view for wider swap spreads in 2014.

# Basel III Regulation and US Rates

**Andrew Hollenhorst**

While a variety of new rules and regulations will come fully into force in 2014 and beyond, the Basel III LCR and SLR directly influence the costs and benefits of holding Treasury securities. We think the effects of the SLR will be felt in 2014 as reduced repo balance sheet and dealer inventories lead to lower Treasury market liquidity, more volatile short-term price movements and more dislocations in the yield curve. While the LCR may eventually be an important driver of demand for Treasuries, we think the full effects will only be felt when the Fed begins to drain reserves from the banking system in 2015 or 2016.

## Basel III SLR

**SLR may be the most important regulatory development in 2014 for rates markets.**

In our view, the supplementary leverage ratio (SLR) is the most important near-term regulatory concern for rates investors. Starting in 2018, the 8 designated global systemically important banks (G-SIBs) in the US will need to achieve at least a 5% ratio of tier-1 capital / assets.[7] Assets included in the denominator have been left intentionally very broad and crucially no risk-weighting scheme is applied. In other words, under a binding SLR a bank would need to raise the same amount of capital to support $1 billion of US Treasury assets as it would $1 billion in high yield corporate debt.

**Banks will look to reduce low return assets like repo.**

While the SLR does not come into effect until 2018 (Figure 76), banks are already releasing pro-forma SLRs and seeking to comply well in advance. Over the last few years US banks have focused on tier-1 / risk weighted asset (RWA) ratios and have maximized return on RWA. In 2014 Citi thinks the SLR will be the binding regulatory constraint for US banks, shifting attention to return on total assets. We expect the 8 G-SIBs to reduce assets that generate relatively low return. In particular, repo books and dealer inventories of Treasuries and high quality assets are likely to shrink.

**May see more short-term volatility and yield curve dislocations.**

**Rates impact:** With dealers less willing to expand balance sheets to absorb client Treasury trading flows we expect to see more volatility in Treasury prices. Historically, dislocations in the yield curve have been arbitraged away by relative value investors who use the repo market to leverage relatively small mispricings into reasonable return on capital. As leverage becomes harder to come by, arbitrageurs will become less active and we expect to see larger and more persistent yield-curve dislocations. We do not expect the SLR to have a first order impact on the level of interest rates or on swap spreads.

Figure 76. SLR does not come into effect until 2018, but banks may comply ahead of schedule

US Implementation of Basel III Regulation

| | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 |
|---|---|---|---|---|---|---|---|
| Tier-1 Common / RWA | | 4% | 4.5% | 4.5% | 4.5% | 4.5% | 4.5% |
| Tier 1 / RWA | | 5.5% | 6% | 6% | 6% | 6% | 6% |
| SLR | | | Begin Reporting | | | 5% | 5% |
| LCR | | Final Rule? | 80% | 90% | 100% | 100% | 100% |

Source: Citi Research

[7] The ratio is 5% at the bank holding company level and 6% at the depository institution level. For the G-SIBs, the 5% BHC ratio is likely to be the binding constraint as the BHC includes the firm's dealer business.

## Basel III Liquidity Coverage Ratio (LCR)

Figure 77. Banks will need to hold HQLA...

| Level | HQLA | Haricut | Limit |
|---|---|---|---|
| 1 | Tsy, Fed Rvs, GNMA | 0% | |
| 2A | Agency MBS and Debt | 15% | 40% |
| 2B | IG Corp | 50% | 15% |

Source: Citi Research

Figure 78. ...to cover potential outflows

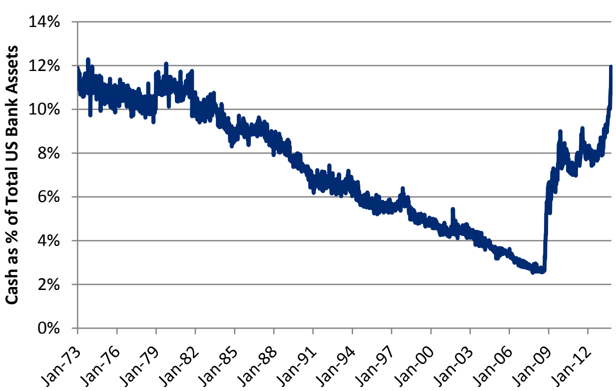
| LCR Outflow Rates | |
|---|---|
| Retail Deposits | 3% |
| Brokered Deposit | 40% |
| Operational Deposit | 25% |
| Non-Fin Corp Deposit | 40% |
| Fin Corp. Deposit | 100% |
| CP | 100% |
| Tsy Repo | 0% |
| MBS Repo | 15% |
| IG Corp Repo | 50% |
| Other Repo | 100% |

Source: Citi Research

The LCR is meant to ensure that US banks hold sufficient high quality liquid assets (HQLA) to satisfy potential cash outflows during a 30-day stress scenario. To form the LCR denominator banks apply hypothetical outflow rates, determined by regulators, to their various liabilities. For instance, retail deposits are considered relatively sticky and only 3% are assumed to run off, whereas 40% of brokered deposits are expected to run-off. (Figure 78) The numerator is meant to reflect total assets that can be easily converted to cash at close to full value, either by outright sales or through repo transactions. Level 1 HQLA includes Treasuries, Fed reserves and GNMA MBS and count at full value in the numerator, whereas Agency debt and MBS (level 2A) receive a 15% haircut and IG corporates and certain equities (level 2B) receive a 50% haircut. (Figure 77) All else equal, the LCR, to the extent it becomes a binding constraint, should richen HQLA like Treasuries relative to other fixed income instruments as well as relative to swaps.

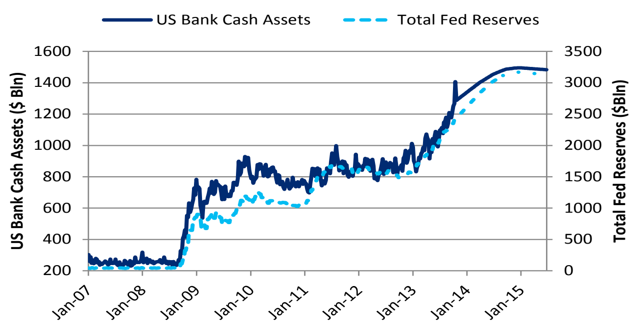
**Rates impact:** Currently, most large banks are unlikely to have trouble meeting 100% LCRs due to elevated cash on the balance sheet. As we have detailed, this is a direct result of Fed reserve creation since the banking system as a whole must hold the total supply of Fed reserves. Given our expectation for continued expansion of reserves in 2014, we think there will be little increased demand for HQLA this year as a consequence of LCR. However, longer term we expect the Fed to withdraw substantial reserves as it exits the long period of accommodation in 2015. At this point LCR concerns are likely to resurface and Treasuries may richen.

Figure 79. US Banks hold largest cash allocation since 1979



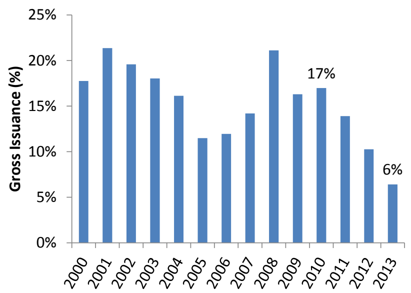

Source: Citi Research, Federal Reserve H.8

Figure 80. Continued reserve creation



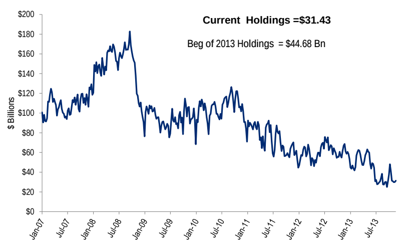

Source: Citi Research, Federal Reserve

Figure 81. Agency Debt as % of Overall Fixed-Income Gross Issuance



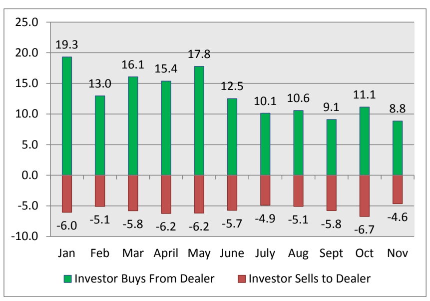

Source: SIFMA

Figure 82. Dealer Holdings of Agencies



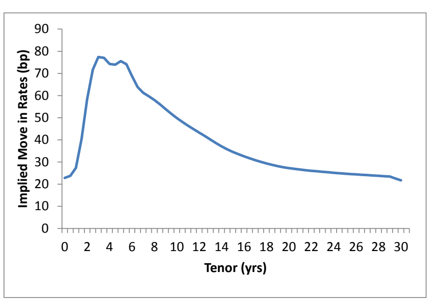

Source: Citi Research, HAVER

Figure 83. Secondary Mkt. Liquidity




Source: Citi Research, TRACE, BBG

Figure 84. Implied Rates Move in 2014




Source: Citi Research, BBG

# 2014 Agency Debt Outlook

## Key Themes for 2014

### Continued Decline in Supply and Size of the Agency Market

Agency debt continues to decline and now accounts for just 6% of overall fixed income issuance[8]. Just three short years ago agencies constituted close to 17% of overall fixed income issuance in the US (Figure 81).

Fannie Mae and Freddie Mac continue to reduce the size of their retained portfolios in keeping with their Treasury preferred stock purchase agreements (PSPA). Fannie Mae was down -$70 billion in net issuance and Freddie was down -$32 billion in 2013. We expect the trend to continue in 2014 and project Fannie to be down -$84 billion and Freddie to be down -$73 billion in overall net issuance (including discount notes).

The Federal Home Loan bank, in contrast, actually increased advances to member banks and ended the year with +$52 billion in net issuance. We believe that the increase in advances was generally led by the larger members on the back of proposed regulations on liquidity ratios. We expect the trend to continue and the home loan banks to end 2014 with positive +$41.7 billion in overall net issuance (including discount notes).

### Changing Liquidity in the Agency Debt Markets

Agency debt is explicitly exempted from the Volker rule's prop-trading clause. This should be a positive for liquidity in the agency debt market compared to other non-exempt sectors like corporates or non-agency mortgages.

On the other hand, liquidity in the agency debt market might suffer from a) the declining supply in agency debt as well as b) if primary dealers continued reduction in the size of their agency debt holdings due to balance sheet constraints (Figure 82).

Figure 83 highlights that dealer sales to investors have been going down while investor sales to dealers have stayed pretty constant over the year. This suggests that the dealer community has actually been intermediating effectively in secondary markets.

Nonetheless, intermediation in the primary market seems to have suffered. New issue concessions have increased in some instances for benchmark deals. In addition, dealers have been less willing to take down larger new issue callable deals without having a strong commitment from investors.

### Imminent Rise in Rates Could Temper Investor Sentiment

Anecdotally, some investors have been very concerned about a sharp sell-off in rates once the Fed starts to taper. Our economists do expect the taper to happen in the first quarter. However, our rates strategists do not expect a repeat of the May/June sell-off. They expect rates to end up very close to implied forwards (Figure 84) for the year which suggest a flattening in the 5s10s curve with 5s rising over 70bp in 2014 while 10s

[8] https://www.sifma.org/WorkArea/DownloadAsset.aspx?id=8589942781

rising by 50bp. While not trivial, forward rates being realized is not enough of a selloff to produce negative returns over the course of the year.[9]

We utilize our spread model and our strategist forecasts to project agency spreads in Figure 85. Our model projects that agency spreads will widen 6bp to 12bp across the curve aside from the 2-year, for which we project 1bp of widening.

Figure 85. Agency Spread Estimates

| Tenor | Current | 1Q, 2014 | 2Q, 2014 | 3Q, 2014 | 4Q, 2014 |
|---|---|---|---|---|---|
| 2yr | 4 | 5 | 5 | 5 | 5 |
| 3yr | 6 | 11 | 12 | 13 | 14 |
| 5yr | 19 | 21 | 24 | 26 | 28 |
| 7yr | 24 | 27 | 31 | 32 | 34 |
| 10yr | 38 | 37 | 43 | 44 | 44 |
| 20yr | 59 | 65 | 68 | 70 | 71 |

Source: Citi Research

## Agency Spreads Resilient in the Face of GSE Headlines

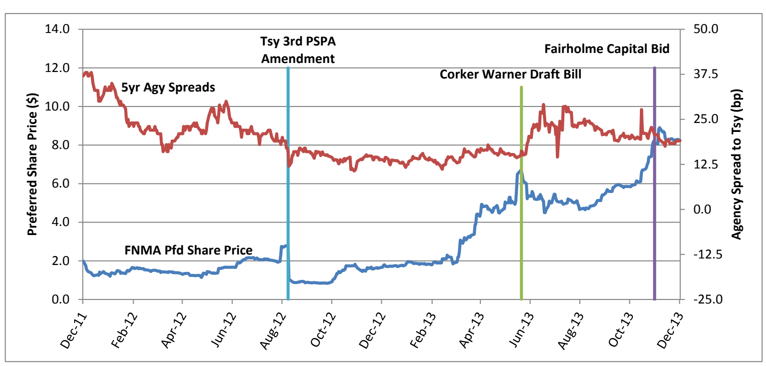
Agency debt spreads have been impressively resistant to GSE headlines in 2013, though preferred and common stock of the agencies has reacted more – declining on the announcement of the 3rd Treasury PSPA and swinging over 10% on the unveiling of the Corker-Warner bill.

There is likely to be continued headline volatility in 2014, but unless the news becomes more concrete and material to agency senior debt, we do not expect it to roil agency markets unduly.

Listed below are some events to be aware of in 2014:

- The progress and evolution of the Corker-Warner Bill (draft introduced in the 2nd quarter – the green line in Figure 86) and the PATH act in the Senate and the House respectively.
- Public pressure from private investors – for example Fairholme capital announcing its desire to acquire the insurance businesses of FNMA and FHLMC (purple line in Figure 86) and Perry capital's lawsuit challenging the 3rd Treasury amendment to the PSPA.

Figure 86. Agency Markets and Some Events




Source: Citi Research

[9] If Implied forwards are realized every point on the rates curve earns the same risk free rate.

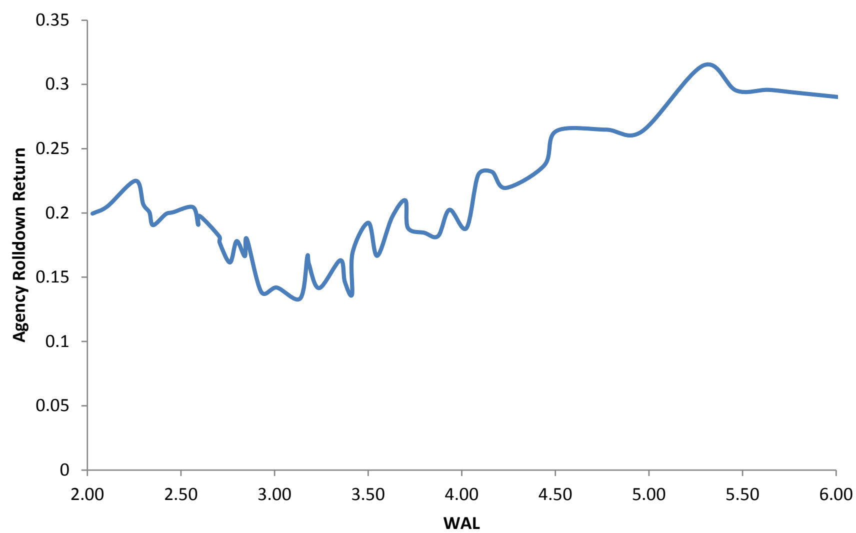
## Recommendations for 2014

### 1. Agency Bullets: Long the 5-year

We find 5-year agency bullets attractive from both a roll-down perspective as well as from a break-even perspective.

In Figure 87 we show projected one-year nominal returns for each bond in the agency index, given implied forwards and our projected agency spread changes. We find the five year point to be the hands on favorite in spite of the treasury curve implying a 70bp sell-off over the year.

Figure 87. 1-year Projected Returns (using implied forwards and our projected spread changes)




Source: Citi Research

5-year agencies also display the widest breakeven spread change versus Treasuries. The breakeven spread change for five-year agencies over the course of the year exceeds our projected spread widening.

Figure 88. 1-year Break-even Changes

| Maturity | 2 | 3 | 5 | 7 | 10 |
|---|---|---|---|---|---|
| 12/05/2013 OAS to Tsy | 4 | 6 | 19 | 24 | 38 |
| 52 wks Spread Range | 0 ~ 10 | 3 ~ 14 | 11 ~ 29 | 15 ~ 41 | 19 ~ 57 |
| | | | | | |
| Modified Breakeven Change Over 1 Year (= 1 + 2) | 7.8 | 5.4 | 11.9 | 7.2 | 10.6 |
| 1. Curve Implied spread change over 1yr | 2.1 | 2.0 | 6.6 | 2.6 | 4.7 |
| 2. OAS constant Breakeven Change (carry advantage) | 5.7 | 3.4 | 5.4 | 4.7 | 5.9 |
| Final Modified BE Spread as a multiple of current spread | 2.6 | 2.4 | 1.6 | 1.4 | 1.3 |
| | | | | | |
| 1yr Spread Volatility (std. deviation over 1 year history) | 1.4 | 2.0 | 4.3 | 6.9 | 10.3 |
| 1yr Modified BE Chg/Volatility | 5.66 | 2.68 | 2.80 | 1.05 | 1.03 |

Source: Citi Research

### 2. Callable Agencies : An Attractive Environment in 2014

The rates market in 2014 is likely to prove friendly for callable agencies:

- Our strategists forecast rates ending the year close to where forwards imply. This is optimal for callable agencies as it mitigates extension risk.

- Our volatility strategist expects volatility to decline over the coming year based on a) rates being much closer to fair value, b) the reduced dispersion in projected economic outcomes across the street and c) the historical behavior of vol. post-crisis (page 34).

- Our volatility strategist also expect the GSEs to continue to source more volatility in the callable market as regulation increases the price of sourcing vol in the derivatives markets.

Currently, callables are somewhat rich compared to like-duration agency bullets on an OAS basis and generally have coupons below forward rates. However, they still offer attractive base-case returns relative to the like-duration agency bullets. We recommend being opportunistic in choosing entry points.

We currently prefer 6-month locks on 4- and 5-year tenors but expect specific structure recommendations to change with the market over the year. They have the least negative OAS to like-duration bullets and are reasonable returns in the base case.

Figure 89. Estimated European Callables and Projected Returns (Pricing: 12/6/13; 2 weeks settle)

| | | | | | | | | Differences Versus Comparable Duration Bullet Return Diff. Versus Changes to Six-month Forward Rate | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Structure | Fair Value Coupon | Eff.Dur | Convex | Vega | Fwd Agy Rate | Yld | OAS Pickup | -75bp | -50bp | -25bp | 0bp | 25bp | 50bp | 75bp | Comparable Duration Bullet |
| 3NC3M | 0.74 | 1.56 | -3.81 | 0.00 | 0.78 | 0.43 | -5 | -0.21 | -0.21 | -0.16 | 0.16 | -0.19 | -0.54 | -0.89 | FNMA 2.375, 07/2015 |
| 3NC6M | 0.76 | 1.88 | -2.83 | 0.00 | 0.84 | 0.38 | -6 | -0.18 | -0.18 | -0.05 | 0.18 | -0.11 | -0.39 | -0.67 | FHLMC 4.750, 11/2015 |
| 3NC1Y | 0.75 | 2.32 | -1.44 | 0.00 | 0.99 | 0.24 | -4 | -0.29 | -0.26 | 0.08 | 0.12 | 0.01 | -0.14 | -0.30 | FHLMC 2.500, 05/2016 |
| 4NC3M | 1.26 | 1.89 | -4.41 | -0.01 | 1.30 | 0.88 | -3 | -0.24 | -0.24 | -0.11 | 0.26 | -0.12 | -0.64 | -1.15 | FHLMC 4.750, 11/2015 |
| 4NC6M | 1.28 | 2.31 | -3.02 | -0.01 | 1.39 | 0.77 | -1 | -0.39 | -0.35 | 0.08 | 0.47 | 0.07 | -0.31 | -0.69 | FHLMC 2.500, 05/2016 |
| 4NC1Y | 1.25 | 2.91 | -1.59 | -0.01 | 1.58 | 0.44 | -8 | -0.68 | -0.17 | 0.17 | 0.21 | 0.08 | -0.13 | -0.37 | FNMA 5.000, 02/2017 |
| 5NC3M | 1.71 | 2.13 | -4.68 | -0.01 | 1.78 | 1.24 | -8 | -0.37 | -0.37 | -0.08 | 0.36 | 0.05 | -0.62 | -1.29 | FNMA 5.000, 03/2016 |
| 5NC6M | 1.76 | 2.61 | -3.14 | -0.02 | 1.87 | 1.15 | -5 | -0.48 | -0.28 | 0.25 | 0.77 | 0.33 | -0.22 | -0.77 | FHLMC 2.000, 08/2016 |
| 5NC1Y | 1.76 | 3.36 | -1.76 | -0.03 | 2.08 | 0.67 | -12 | -0.59 | -0.04 | 0.31 | 0.37 | 0.24 | -0.03 | -0.35 | FHLMC 5.500, 08/2017 |
| 6NC6M | 2.21 | 2.90 | -3.23 | -0.03 | 2.35 | 1.40 | -9 | -0.80 | -0.21 | 0.38 | 0.99 | 0.71 | 0.01 | -0.67 | FNMA 5.000, 02/2017 |
| 6NC1Y | 2.22 | 3.76 | -1.90 | -0.04 | 2.57 | 1.04 | -9 | -0.56 | 0.03 | 0.42 | 0.51 | 0.39 | 0.13 | -0.22 | FNMA 0.875, 10/2017 |
| 7NC3M | 2.52 | 2.64 | -5.21 | -0.03 | 2.64 | 1.84 | -6 | -0.89 | -0.55 | -0.02 | 0.55 | 1.13 | -0.45 | -1.40 | FHLMC 5.125, 10/2016 |
| 7NC6M | 2.56 | 3.17 | -3.40 | -0.04 | 2.75 | 1.62 | -13 | -0.82 | -0.17 | 0.49 | 1.15 | 1.04 | 0.19 | -0.64 | FNMA 5.000, 05/2017 |
| 7NC1Y | 2.57 | 4.13 | -2.09 | -0.06 | 2.97 | 1.19 | -14 | -0.57 | 0.07 | 0.48 | 0.58 | 0.48 | 0.19 | -0.23 | FHLMC 0.875, 03/2018 |
| 8NC6M | 2.83 | 3.44 | -3.71 | -0.06 | 3.07 | 1.81 | -13 | -0.94 | -0.19 | 0.55 | 1.29 | 1.30 | 0.33 | -0.62 | FHLMC 1.000, 06/2017 |
| 8NC1Y | 2.84 | 4.51 | -2.36 | -0.08 | 3.29 | 1.22 | -17 | -0.68 | 0.02 | 0.49 | 0.63 | 0.53 | 0.21 | -0.24 | FNMA 1.875, 09/2018 |
| 10NC3M | 3.19 | 3.36 | -6.86 | -0.05 | 3.49 | 2.10 | -13 | -1.46 | -0.76 | -0.05 | 0.70 | 1.45 | -0.53 | -1.84 | FHLMC 5.500, 08/2017 |
| 10NC6M | 3.30 | 3.94 | -4.41 | -0.08 | 3.59 | 2.05 | -8 | -1.09 | -0.21 | 0.66 | 1.52 | 1.70 | 0.49 | -0.69 | FNMA 0.875, 12/2017 |
| 10NC1Y | 3.43 | 5.17 | -2.89 | -0.10 | 3.80 | 1.50 | 0 | -0.75 | 0.09 | 0.66 | 0.84 | 0.71 | 0.33 | -0.21 | FHLMC 1.750, 05/2019 |

Source: Citi Research

# Supply Projections for 2014

We project that the largest three agencies will have a total negative net issuance in 2014 of -$94 billion. Approximately -$60 billion will be in term debt and another -$34 billion will be in discount notes (Figure 90). This is a larger projected decline than the -$50 billion in total net issuance the agencies have seen so far in 2013, which was impacted by two major events:

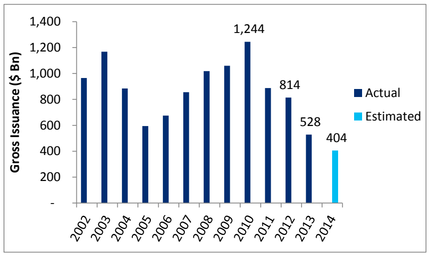
> a) The one-time accounting changes at Fannie and Freddie that increased the cash dividend that the GSEs paid the Treasury. They tapped debt markets to raise cash for these payments.
>
> b) A surge in advances by large FHLB members looking to potentially strengthen their liquidity position under newly proposed LCR regulation.

Figure 90. Supply Projections 2014

| | Total | Fannie Mae | Freddie Mac | FHLB |
|---|---|---|---|---|
| Discount Notes Net Issuance | -33,965 | -10,672 | -44,887 | 21,594 |
| Term Debt Net Issuance | -60,444 | -73,426 | -28,772 | 41,754 |
| Term Debt Gross Issuance | 404,154 | 57,094 | 85,976 | 261,084 |

Source: Citi Research

Figure 91.Gross Issuance of Agencies

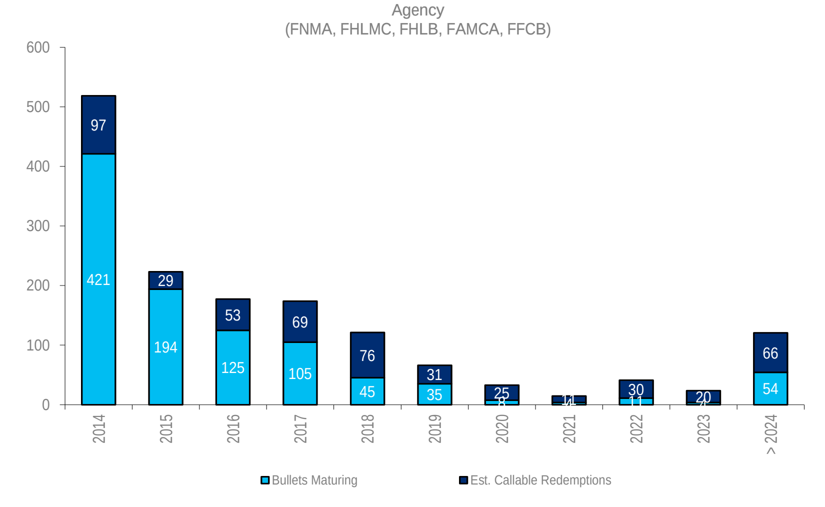



Source: Citi Research

We estimate that 2014 will be the lowest gross issuance year for the agencies for at least a decade (Figure 91). In the wake of the large rise in rates in 2013, most callable agencies have extended and will reduce the need for the GSEs to roll over debt. For the full year, assuming forwards are realized; we expect only $84 million of callable agencies to be redeemed.

In Figure 92 we show our projected redemption profile of current debt, assuming that callable agencies that aren't redeemed in 2013 will extend to their maturities. This extended callable debt increases the amount of expected redemptions substantially at longer tenors. Indeed extended callable agencies could account for the vast majority of current agency debt coming due in 2018 and later (Figure 92).

Figure 92. Estimated Bullet & Callable Redemptions, Implied Forwards Scenario



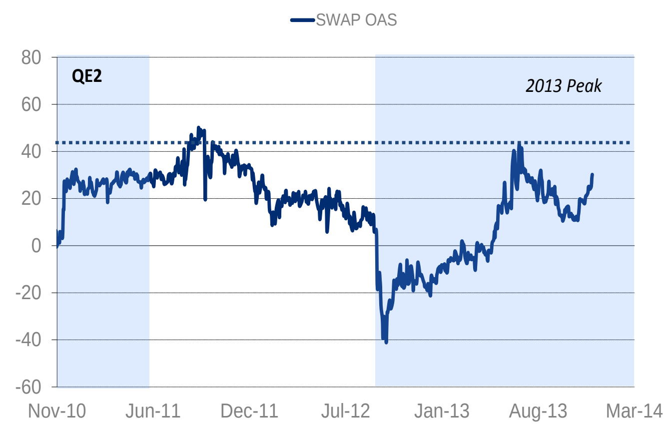

Source: Citi Research

# Agency MBS Outlook for 2014

**Ankur Mehta**

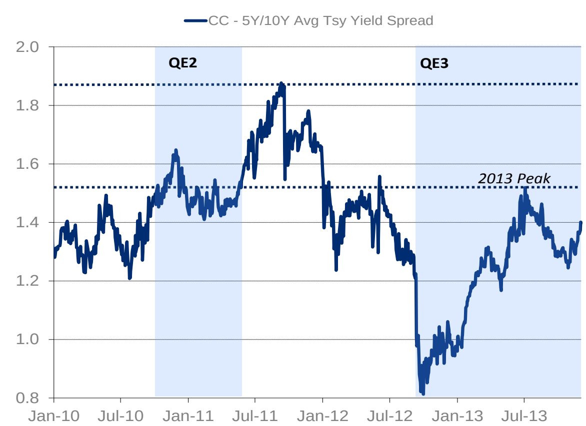
Agency MBS investors entered 2013 holding a view that technicals will be positive due to the sizeable monthly purchases of the Fed and spreads will hold up relatively well. However, the selloff triggered by the stronger than expected May NFP print caught the market off-guard. The first leg of the widening was due to changing expectations of the Fed's tapering of their QE program. Then, as rates sold off and spreads widened, mortgage REITs were forced to de-lever and money managers started to see redemptions. This led to production coupon spreads reaching their widest levels since 2012 (Figure 93 and Figure 94). As a result, pretty much every investor type was a net seller of agency MBS in 2013, with the Fed being the only buyer.

Figure 93. Production coupon LOAS




Source: Citi Research, Yiedlbook

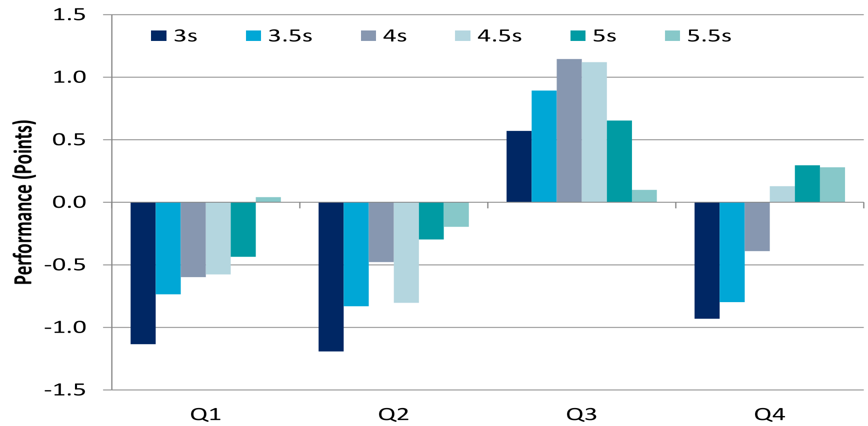
Figure 94. Current coupon yield pickup versus 5s and 10s




Source: Citi Research, Bloomberg

As market expectations of QE evolved through the year, lower coupons bore the brunt of the blow and underperformed significantly in Q1, Q2 and Q4 of 2013 (Figure 95). Overall, FN 3.0s underperformed their duration hedges by just over 2.5 points while FN 5.0s-6.0s outperformed by 0.25 to 0.5 point as the slowdown in HARP speeds and generally bearish view on rates drove investors into the latter.

Figure 95. Lower coupons underperformed significantly in 2013




Source: Citi Research, Yieldbook

# Major Themes for 2014

Going into 2014, we think the following themes will dominate the agency MBS market:

- **Technicals likely to be modestly unfavorable:** We expect net supply in agency MBS to be $235bn in 2014 as the increase in home sales is offset by more whole loan retention by banks. From the demand side, we think the biggest change could come from China, which has become a net buyer of agency MBS recently. We expect overseas investors to be net buyers of $10bn next year. Bank holdings of agency MBS are likely to be driven by regulation and we expect their holdings to increase by $10bn. Mortgage REITs are likely to focus on preservation of equity rather than dividend enhancement and we expect their holdings to decline by approximately $10bn. The GSEs have shown willingness to add MBS when spreads widen meaningfully and they could provide an unexpected source of demand when the Fed tapers. We expect their holdings to decline by $50bn next year. We expect the Fed to start tapering their purchases in March and end QE3 in October of 2014. Based on this schedule, they are likely to be a net buyer of $300bn agency MBS in 2014 and will end up owning a third of the fixed-rate agency MBS outstanding. For supply and demand to balance out, the agency MBS holdings of fixed-income funds would need to decline by $15bn over the next year, which would imply that they will move from a 1% overweight to neutral on the sector. Overall, we think technicals are modestly unfavorable.

- **Policy risk will remain elevated:** Capacity constraints have eased significantly since the start of 2013 and are unlikely to arise in the foreseeable future, so competition should remain robust in 2014. Continued MSR transfers could also continue to boost competition. Housing fundamentals should continue to improve, but at a slower pace than in 2013. Amid this backdrop, we expect HARP speeds to slow only slightly through the first half of 2014 if rates stay here. In non-HARP space, the market will continue to focus attention on turnover speeds in 2014. On the policy front, HARP expansion uncertainty could persist for a while. But as we have been highlighting since late-2012, we continue to think re-introducing the streamline refinance program is a potentially less contentious and more impactful method of increasing refinancings. Based on the pricing of several risk sharing transactions by Fannie Mae and Freddie Mac, we estimate that the fair value of the g-fee is close to the current level, so further significant increases are unlikely.

- **Outlook for GNMAs:** We expect net issuance in the sector to increase modestly from $110bn to $120bn in 2013. We expect GNMA holdings of domestic banks to decline further as they are likely to meet their LCR needs through an increase in reserves and cash on balance sheet. Overseas investors are likely to be modest buyers whereas holdings of money managers are likely to remain unchanged. On the prepay front, FHA-to-conventional refinancings will continue to dominate prepays. We recommend shorting the G2/FN 3.5s swap at current levels.

- **Positioning in the agency MBS market:** We think that MBS investors are much better positioned for a selloff in rates than they were in May of 2013. This argues that spreads should not widen to levels seen at that time. However, carry on MBS is now much lower than it was at that time making mortgages less attractive. Overall, we think that MBS has further room to widen given the lower carry and absence of yield based buyers in the short term and thus recommend a modest underweight. Across the coupon stack, we think FN 4.5s are priced attractively.

We think specified pool payups will decline further and recommend staying long TBAs. Within specs, LLB 4.0s are looking attractively priced.

## Supply and Demand Projections for 2014

Over the last year, every major private investor type reduced their holdings of agency MBS with the Fed being the only investor adding bonds. As we enter 2014, the biggest question in the minds of investors is which investor type will provide the bid for mortgages once the Fed ends its QE program. We think that this demand is likely to come from domestic banks and overseas investors. We discuss supply and demand technicals in more detail in the following section. Our key takeaways are as follows:

- **Expect net supply of $235bn:** We expect net issuance in the sector to be $235bn and gross issuance to be between $1,020 to 1,050bn if rates stay here. Although we expect home sales to increase modestly, we expect agency securitization rates will decline as banks retain more whole loans and non-agency securitization picks up modestly. We also anticipate some net supply from the securitization of reperformers and modified loans in GSE portfolios.

- **Regulation to drive bank demand:** The various regulation that domestic banks are faced with makes investing a portfolio optimization problem with LCRs, SLRs, Tier-1 capital and AOCI requirements as major drivers. We expect SLRs to be the binding constraint for large banks and LCR to be a binding constraint for some small banks. Overall, we think domestic banks will add $10bn of agency MBS in 2014. Over the longer run, as the Fed starts to tighten and cash on bank balance sheets decline, domestic banks are likely to be forced to buy agency MBS to maintain liquidity (from an LCR perspective) in their portfolio.

- **Overseas investors likely to add $10bn:** China's allocation to agency MBS (versus Treasuries) has declined from 41% in the middle of 2008 to less than 18%. The fact that China has become a net buyer of agency MBS recently is suggesting that they may look to increase their allocation of agency MBS again. Japan, which has been adding Treasuries over MBS, could also become a net buyer once the Fed starts to taper and spreads widen. Overall, we expect overseas investors to be net buyers of $10bn agency MBS in 2014, which will be a significant reversal from 2013 where they are expected to have reduced their holdings by $40bn.

- **REITs to reduce holdings by $10-15bn:** Agency MBS REITs have become more risk averse due to the uncertainty around Fed tapering and are likely to be focused on preserving equity rather than enhancing dividends. Further, repo availability may be adversely impacted by the supplemental leverage ratios and the potential repo surcharge for banks. Overall, we think that REIT holdings of agency MBS will decline by $10-15bn in the next year.

- **Fed likely to add $300bn:** We expect the Fed to start tapering their purchases in March and end QE3 in October of 2014. Based on this schedule, we expect the Fed to be a net buyer of $300bn agency MBS in 2014 and will own a third of the fixed-rate agency MBS outstanding.

- **GSEs likely to reduce holdings by $50bn:** Both GSEs are well below their portfolio limits for 2013, and close to the limits for the end of 2014. We expect their agency portfolios to decline by $50bn in 2014. If spreads widen materially, we think Freddie Mac may even come in and buy agency MBS.

- **Technicals are modestly unfavorable:** For supply and demand to balance out, the agency MBS holdings of fixed-income funds would need to decline by $15bn over the next year, which would imply that they will move from a 1% overweight to neutral on the sector. Overall, we think technicals are modestly unfavorable.

Figure 96 shows the level of agency MBS holdings across different investors as of December 2012 and our estimated holdings as of year end 2013. We also show our expectations of how these holdings will change over the course of the next year. Note that the holdings of "Other Fixed-Income Funds" reflect how much their holdings need to change to balance out supply/demand. We estimate that at the end of the year, $500bn will be held by fixed-income funds, most of which are benchmarked off the fixed-income aggregate indices. For supply and demand to balance out, the agency MBS holdings of these fixed-income funds would need to decline by $15bn over the next year. If this were to happen, money managers would need to marginally reduce their MBS weighting from 32% to 31%. Put another way, money managers will need to go from being 1% overweight to being neutral[10]. This indicates that supply and demand technicals at current spread levels are modestly unfavorable.

Figure 96. Technicals are modestly unfavorable

| Investor Type | Holdings as of Dec'12 ($bn) | Estimated Holdings as of Dec'13 ($bn) | 2013 Change ($bn) | Expected Holdings as of Dec'14 ($bn) | Expected 1-Yr Change ($bn) |
|---|---|---|---|---|---|
| GSEs | 516 | 435 | -81 | 385 | -50 |
| Federal Reserve/Tsy | 926 | 1,580 | 654 | 1,880 | 300 |
| Banks | 1,415 | 1,365 | -50 | 1,375 | 10 |
| Foreign | 680 | 640 | -40 | 650 | 10 |
| REITs | 355 | 270 | -85 | 260 | -10 |
| Agency MBS Funds | 90 | 70 | -20 | 70 | 0 |
| Other Fixed-Income Funds | 579 | 500 | -79 | 485 | -15 |
| Implied MBS Weighting of FI Funds | 36% | 32% | | 31% | |
| Others (Insurance/pension/ State/Local Govts, etc.) | 660 | 640 | -20 | 630 | -10 |
| Outstanding MBS | 5,240 | 5,500 | 260 | 5,735 | 235 |

Source: Citi Research

## Prepayment Outlook for 2014

Rising competition for mortgage lending was one of the key themes in the mortgage market in 2013. Record gain on sale going into 2013 led to rising capacity through the first half of the year. MSR transfers from larger entities (mostly large banks) to smaller entities (mostly non-banks) also helped to increase lending competition. As capacity constraints diminished, and as housing fundamentals became less of a bottleneck to mortgage lending, the non-HARP refi response became less dependent on factors such as LTV and FICO and more dependent on rate incentive. The easing of capacity constraints helped to support fast speeds in the first half of the year. But the dramatic rates selloff in the second half of the year caused the market to focus attention on discount speeds.

Going into 2014, we expect fundamentals to continue to improve, but at a slower pace than in 2013. Mortgage lending standards should continue to ease and private

[10] We assume that agency MBS comprise of approximately 32% of the US Fixed Income Index currently and will decline to 30-31% by the end of next year.

mortgage insurance should also become more readily available. Amid this fundamentals backdrop, our expectations on conventional prepayments and their drivers are summarized below:

- **Capacity constraints unlikely to arise:** The decline in gain on sale and shortening of origination lags suggest that capacity is much less constrained than it was at the start of 2013. Moreover, we do not foresee capacity constraints arising in the foreseeable future. The rates selloff has dramatically reduced demand. Ongoing job cuts also imply that there may still be excess capacity in the system. The easing of lending standards, as reflected by lower FICOs, higher LTVs and higher DTIs on new originations, also suggests that lenders are aggressively competing for origination volume. Continued MSR transfers could also continue to boost competition next year. Many of these acquisitions are focused on HARP collateral which remains in the money. Therefore, these smaller lenders are less likely to cut staff, helping to stabilize industry employment levels.

- **HARP speeds should slow only slightly through 1H 2014:** Aggregate HARP speeds appear to have peaked back in the spring and have since been trending lower. Speeds on FN 2006–2008 vintage 30-year fixed-rate pools fell to 36% CPR in October from 48% CPR back in April/May. Our analysis suggests that the recent slowdown was primarily driven by the rate effect rather than by burnout. Looking ahead, we expect HARP speeds to slow by just a few CPR through the first half of 2014, to around 34% CPR by March 2014 and to 32% CPR by June if rates stay here.

- **Focus on conventional discount speeds:** With more than half of the conventional universe out of the money at current rate levels, the market will continue to focus attention on turnover in 2014. In our base scenario, which assumes moderate HPA and existing home sales growth, we expect speeds on conventional 3.0s of 2012–13 to range from 5.5–6.5% in 2014 and 2015, and speeds on 3.5s of 2011–13 to range from 6.0–7.0%. Under the optimistic scenario, which assumes more aggressive housing market growth in 2014 and 2015, we expect speeds on 3.0s of 2012–13 to range from 6.0–7.25% in 2014 and 2015, and speeds on 3.5s of 2011–13 to range from 6.5–7.7%.

- **HARP expansion uncertainty could persist for a while:** The increased likelihood of a Mel Watt-led FHFA has reignited HARP expansion concerns. If confirmed, we expect him to prioritize principal forgiveness over HARP expansion, although he may be more amenable to expanding HARP if rates rally from here. Regardless, his confirmation will effectively worsen the negative convexity in non-HARP pools, as the likelihood of a change in underwriting is much higher under him than it has been under Ed DeMarco. But as we have been highlighting since late-2012, we continue to think re-introducing the streamline refinance program is a potentially less contentious and more impactful method of increasing refinancings. Separately, he could also reduce LLPAs, loosen underwriting and keep the g-fee and loan limits unchanged.

- **G-fee unlikely to see significant increase:** Based on the pricing of several risk sharing transactions by Fannie Mae and Freddie Mac, we estimate that the fair value of the g-fee is 47 to 53bp, which is close to the current level. Therefore, we think the g-fee will be increased by at most 10bp in the short-term and could even remain unchanged if further risk sharing transactions are priced as competitively as the recent deals.

# Positioning in Agency MBS

Now that we have presented our outlook for technicals and prepayments, we take a closer look at valuations and suggest recommended positioning in the market for 2014. We recommend going into 2014 with a modest underweight on MBS versus Treasuries. Across the coupon stack, we think FN 4.5s are priced attractively. We think specified pool payups will decline further and recommend staying long TBAs. Within specs, LLB 4.0s are looking attractively priced.

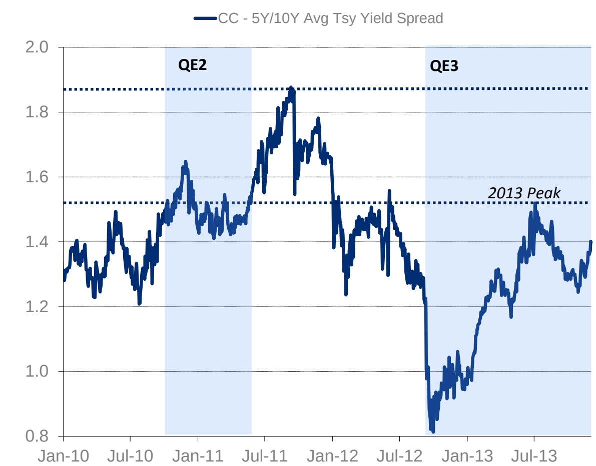
## Spreads have widened, but carry remains low

Agency MBS have been widening since the beginning of November as the much stronger than expected NFP print forced investors to re-assess their QE tapering views. In spite of the widening, production coupon LOAS is still 10-15bp tighter than the wides observed in July of 2013. Similarly, current coupon yield pickup versus the average of the 5-year and 10-year Treasury yields is approximately 10-15pb tighter than the wides in July.

Figure 97. Production coupon OAS is 10-15bp tighter than 2013 wides


Source: Citi Research

Figure 98. Current coupon yield pickup versus 5s/10s is also 10-15bp tighter than the wides seen in July



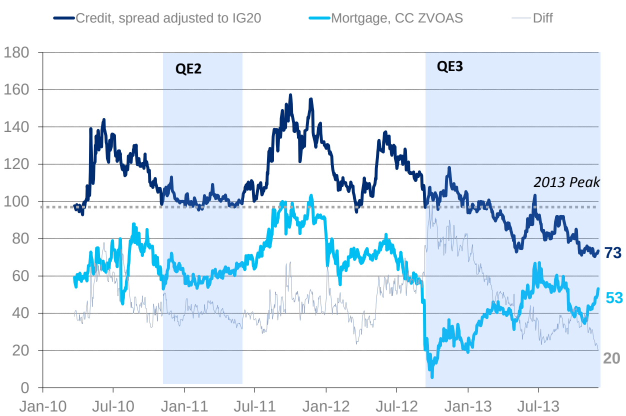

Source: Citi Research

Although spreads have widened recently, carry on MBS continues to remain very low (Figure 99). Hedge adjusted carry on production coupon has declined to just under 1.5 ticks per month as the carry and roll down on duration hedges has exceeded the appreciation in dollar rolls into the selloff. From a money manager perspective, MBS are now looking attractively priced versus corporates (Figure 100). Corporates have historically picked up approximately 40bp of spread over mortgages but are now picking up only 20bp of spread. However, with the economy recovering and the risk being tilted towards an end to the Fed's agency MBS bond buying program, it is not clear if money managers would like to increase their allocation of agency MBS from here. Further, even though corporates offer less spread pick up, they offer carry through roll down which MBS do not offer. Thus, we think money managers will prefer Treasuries and corporates to MBS even at current spread levels.

Figure 99. Production coupon curve hedged carry has declined meaningfully


Source: Citi Research

Figure 100. Corporates are looking tight to MBS




Source: Citi Research

Looking forward, we think the following factors are a positive for the basis:

- **China could start buying agency MBS:** As we have highlighted before, China has started to add MBS to their portfolio in 2013. We think that their purchases could become sizeable if MBS widen approximately 10bp from here.
- **Domestic banks could start adding MBS:** Domestic banks are sitting on record amounts of cash and as yields move higher and spreads widen, they could start to come in and buy. Similar to overseas investors, we think bid from banks is likely to increase as the 10-year reaches 3% and MBS spreads widen 10bp.
- **GSEs could add opportunistically:** Freddie Mac added agency MBS to their portfolio when OAS on production coupons was 10bp wider than current levels. We think they could come in and buy again if spreads widen to similar levels.

On the other hand, the following factors are likely to weigh on the basis:

- **Net issuance continues to be high:** Net issuance has been high and we expect that it will continue to remain high next year. As the Fed starts to taper, the higher net issuance and lower purchases by the Fed could weigh on the basis.
- **Money managers could see further redemptions:** If rates selloff violently, money managers could continue to see redemptions which would force them to sell MBS.
- **REITs could delever further:** Mortgage REITs could de-lever further if rates selloff and spreads widen.

## MBS Basis: Modest underweight

We think that the MBS investor community is much better positioned for a selloff in rates than it was in May 2013. This argues that spreads should not widen to levels seen at that time. However, carry on MBS is now much lower than it was at that time making it a less attractive investment. Overall, we think that MBS have further room to widen given the lower carry and absence of yield based buyers in the short term. However, we do think that overseas investors and domestic banks will come in and start buying MBS if MBS spreads widen 10-15bp from here.

## Outlook for the GNMA Market

The GNMA market has become an increasingly important part of the agency MBS market as it now constitutes of a little over 20% of all agency MBS outstanding. In this article, we look at recent trends in the sector and summarize our views going forward. Our key takeaways are as follows:

- **Net issuance likely to increase modestly:** The GNMA market has seen tremendous growth with outstanding balance tripling from $400bn in 2006 to $1.3trillion in 2013. We think that net issuance in the GNMA sector will increase to $120bn in 2014 as the increase in home sales dominates the reduction in issuance due to FHA to conventional refinancings at higher rate levels.

- **Banks likely to further reduce holdings:** We expect GNMA holdings of domestic banks to decline further as they are likely to meet their LCR needs through an increase in reserves and cash on balance sheet. Overseas investors are likely to be modest buyers whereas holdings of money manageres are likely to remain unchanged.

- **Focus on Ginnie discount speeds:** We think that GN 3.0s will prepay in line with conventional 3.0s whereas GN 3.5s will prepay approximately 1% CPR faster.

- **FHA-to-conventional refis should continue to pick up:** The pace of FHA-to-conventional refis trended higher in 2013 and we expect this trend to continue in 2014. Positive HPA in 2014 should continue to increase the share of FHA borrowers who would find it economical to refinance into a conventional loan. As LTV declines, FICO score progressively becomes less of a limiting factor.

- **Recommended positioning in GNMAs**: We recommend investors short the G2/FN 3.5s swap. This swap has appreciated to around 1-10 recently and we would look to return to a neutral stance on it if it reaches 1point.

## Issuance Outlook

After declining by just over $200bn from 2000 to 2006, the Ginnie Mae sector has seen rapid growth since the credit crisis and the outstanding balance of these securities has increased by over $950bn in the last six years. Although the sector continues to grow, the rate of increase has slowed down with net issuance declining from $233bn in 2009 to $105bn in 2012. We estimate that net issuance in 2013 will be approximately $110bn. Looking into next year, the following two competing effects should drive net issuance in GNMAs:

- **Home sales should increase:** Both existing and new home sales are likely to increase next year. This should lead to an increase in net issuance for GNMAs next year.

- **Reduction in number of FHA to conventional refinancings:** As home prices appreciate, more FHA borrowers will have an incentive to refinance into conventional loans thus reducing net issuance in GNMAs. Assuming home price appreciation is the same, the rate of these refinancings will depend on the level of mortgage rates. The lower the mortgage rates, the larger the propensity of FHA borrowers to refinance into conventional loans and the lower the net

Figure 101. Issuance projections for GNMA MBS

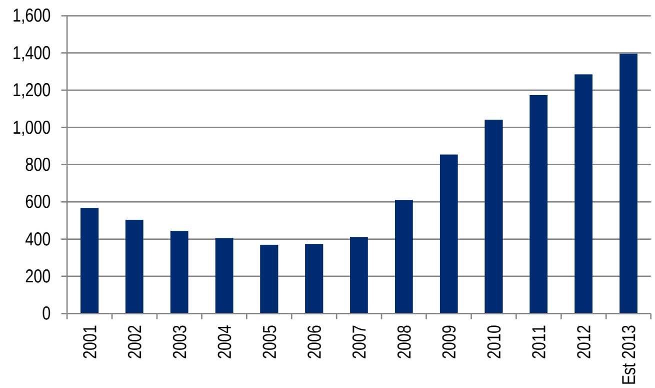
| Year | Gross Iss ($BN) | Net Iss ($BN) |
|---|---|---|
| 2006 | 76 | 3 |
| 2007 | 93 | 32 |
| 2008 | 265 | 192 |
| 2009 | 445 | 233 |
| 2010 | 383 | 192 |
| 2011 | 309 | 141 |
| 2012 | 398 | 100 |
| 2013 (est) | 400 | 110 |
| 2014 (proj) | 310 | 120 |

Source: Citi Research, GNMA, CPR&CDR

issuance. We think rates will trend higher next year which should reduce FHA to conventional refinancings.

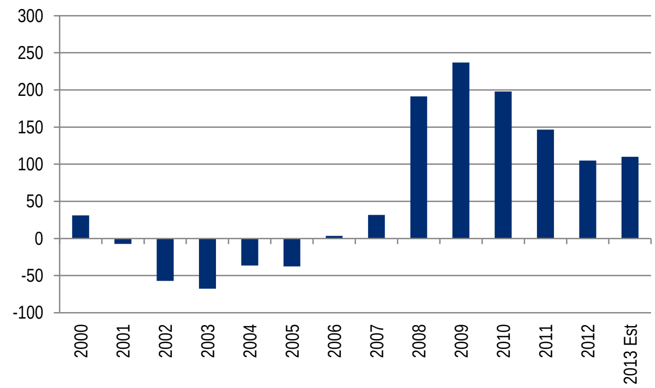
Overall, we think that net issuance in the GNMA sector will increase to $120bn next year as the increase in home sales dominates the reduction in issuance due to FHA to conventional refinancings at higher rate levels.

Figure 102. Ginnie Mae MBS universe has been expanding...




Source: Ginnie Mae, CPR CDR, Citi Research

Figure 103. ... at a declining rate though




Source: Ginnie Mae, CPR CDR, Citi Research

## Demand Outlook for GNMAs

Figure 104 shows the ownership of GNMA MBS across various investor types at different points of time and our estimates for holdings as of September 2013. The GNMA MBS market has more than doubled from $596bn to $1,270bn from the end of 2008 to the end of 2012. Almost half of this increase in net issuance was absorbed by domestic banks whose estimated holdings increased from $187bn to $449bn during this time. Apart from banks, overseas investors were also fairly active in the GNMA market adding approximately $190bn between 2008 and 2012. Money managers in general and Ginnie Mae funds in particular provided the third major bid for GNs adding $79bn to their holdings during this period. More recently, in 2013, all of these investor types have reduced their holdings of GNMA MBS with the Fed absorbing all of the supply coming into the market.

Figure 104. The Fed has been the only buyer in 2013

| | | | | Change | |
|---|---|---|---|---|---|
| Investor Type | Dec'2008 | Dec'2012 | Sep'2013 | 2008-2012 | YTD |
| Federal Reserve | 0 | 142 | 270 | 142 | 128 |
| Banks | 187 | 449 | 400 | 262 | -49 |
| Foreign | 120 | 310 | 300 | 190 | -10 |
| GNMA Funds | 39 | 70 | 55 | 31 | -15 |
| Other Govt. Funds | 32 | 80 | 78 | 48 | -2 |
| Others (MMs + Ins. Companies + State/Local Govts) | 218 | 219 | 217 | 1.2 | -2 |
| Outstanding Balance | 596 | 1270 | 1320 | 674 | 50 |

Source: Citi Research, Federal Reserve, TIC

# Appendix A-1

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